

Linear Parameter Varying $\mathcal{H}_\infty$ Synthesis for Active Suspension of a Quarter-Car

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Abstract

This study presents the design of a Linear Parameter Varying (LPV) $\mathcal{H}_\infty$ controller for an active quarter-car suspension system to optimize the critical trade-off between passenger comfort and road handling. Evaluated across standard ISO 8608 road profiles, the gain-scheduled $\mathcal{H}_\infty$ approach is benchmarked against passive and Linear Quadratic Regulator (LQR) systems. Simulation results demonstrate that the proposed controller increases passenger comfort and helps road handling by minimizing tire lift-off, maintaining robust performance and stability under varying road roughness conditions.

Keywords— LPV, $\mathcal{H}_\infty$, active suspension, quarter car

Contents

1	Introduction	6
2	Literature Review	7
3	Work	7
3.1	Quarter-car Model	7
3.2	Road Profile	9
3.3	LQR Design	9
3.4	$\mathcal{H}_\infty$ Synthesis	10
3.5	Methodology	12
4	Results	13
4.1	Baseline Analysis	13
4.2	Road Profile - Speed Matching	14
4.3	Preliminary $\mathcal{H}_\infty$ Synthesis	14
4.4	Time-Domain Comparisons	14
4.5	β -Sweep	16
4.6	Cross Analysis	18
5	Conclusions and Future Work	22
6	Appendix	24
6.1	Figures	24
6.2	Codes	56

List of Figures

1	Bode plot of Wab filter.	11
2	Bode plot comparison between open-loop and LQR augmented systems. .	13
3	Bode plot evaluation for the $\mathcal{H}_\infty$ controller synthesized for Road Type A.	15
4	Time-domain simulation comparison of open-loop, LQR, and $\mathcal{H}_\infty$ strategies evaluated on Road Type D.	15
5	$\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type A.	24
6	$\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type B.	25
7	$\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type C.	26
8	$\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type D.	27
9	$\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type E.	28
10	$\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type F.	29
11	$\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type G.	30
12	$\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type H.	31
13	Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road A.	32
14	Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road B.	33
15	Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road C.	34
16	Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road D.	35
17	Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road E.	36
18	Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road F.	37
19	Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road G.	38
20	Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road H.	39
21	Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road A.	40
22	Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road B.	41
23	Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road C.	42
24	Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road D.	43

25	Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road E.	44
26	Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road F.	45
27	Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road G.	46
28	Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road H.	47
29	Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road A.	48
30	Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road B.	49
31	Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road C.	50
32	Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road D.	51
33	Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road E.	52
34	Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road F.	53
35	Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road G.	54
36	Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road H.	55

List of Tables

1	Nominal ('nom') Quarter-Car Model Parameters	8
2	Road surface classification and ISO8608 PSD parameters	9
3	Maximum admissible speeds and road roughness limits determined per ISO 8608 class.	14
4	RMS performance of the $\mathcal{H}_\infty$ β -sweep controllers (a_b : body acceleration [m/s ²], t_d : tire deflection [cm])	17
5	Cross-test body-acceleration RMS [m/s ²], $\beta = 0.05$ (rows: design road, cols: test road)	18
6	Cross-test tire-deflection RMS [cm], $\beta = 0.05$ (rows: design road, cols: test road)	19
7	Cross-test suspension-deflection (rattle space) RMS [cm], $\beta = 0.05$ (rows: design road, cols: test road)	19
8	Cross-test body-acceleration RMS [m/s ²], $\beta = 0.50$ (rows: design road, cols: test road)	19
9	Cross-test tire-deflection RMS [cm], $\beta = 0.50$ (rows: design road, cols: test road)	20
10	Cross-test suspension-deflection (rattle space) RMS [cm], $\beta = 0.50$ (rows: design road, cols: test road)	20
11	Cross-test body-acceleration RMS [m/s ²], $\beta = 0.95$ (rows: design road, cols: test road)	20
12	Cross-test tire-deflection RMS [cm], $\beta = 0.95$ (rows: design road, cols: test road)	21
13	Cross-test suspension-deflection (rattle space) RMS [cm], $\beta = 0.95$ (rows: design road, cols: test road)	21

1 Introduction

Car suspension systems play a pivotal role in vehicle design by managing the fundamental trade-off between passenger ride comfort and road handling [4, 5]. Ride comfort is heavily dependent on isolating the vehicle chassis from road disturbances, particularly in the 4–8 Hz frequency range where the human body is most sensitive to vertical vibrations. Conversely, road handling requires maintaining firm and continuous contact between the tires and the road surface to ensure maneuverability and safety. While conventional passive suspensions are fundamentally limited by fixed mechanical spring and damper settings, active suspension systems overcome these limitations by introducing controlled energy into the system, thereby offering the potential to simultaneously improve both performance metrics [5].

In automotive active suspension modeling, assuming zero tire damping leads to a significant theoretical flaw: it falsely implies that at the wheel-hop frequency, the motions of the sprung and unsprung masses become completely decoupled. Consequently, it creates the misleading conclusion that the vertical acceleration of the sprung mass will remain unaffected, regardless of the forces exerted between them [1].

Historically, optimal state-feedback strategies such as the Linear Quadratic Regulator (LQR) have been widely employed as standard benchmarks for active suspension systems [2, 4]. However, traditional LQR design relies heavily on time-domain weight tuning and often struggles to directly address frequency-specific performance requirements. In contrast, $\mathcal{H}_\infty$ synthesis has emerged as a highly effective robust control strategy for vehicle dynamics [6, 9]. The primary motivations for utilizing $\mathcal{H}_\infty$ synthesis for this specific problem are twofold:

- **Intuitive Frequency-Domain Design:** The requirements for passenger comfort and road holding are best described by frequency-domain bounds. $\mathcal{H}_\infty$ synthesis allows engineers to use explicit frequency-dependent weighting filters to directly shape the closed-loop performance, making the design process highly intuitive.
- **Robust Stability Guarantees:** $\mathcal{H}_\infty$ synthesis provides strict mathematical guarantees that the worst-case energy gain (the $\mathcal{H}_\infty$ norm, γ) from exogenous disturbances (such as road profiles) to the performance outputs will be minimized. This bounded transfer function is critical for guaranteeing stability under harsh road conditions.

Furthermore, vehicle dynamics are inherently non-stationary; operating conditions change constantly due to varying vehicle speeds, shifting road roughness profiles, and suspension deflection limits. To accommodate these variations without sacrificing robust stability, Linear Parameter Varying (LPV) control techniques are highly effective. LPV allows for dynamic gain-scheduling based on measurable system parameters in real-time [7, 10].

Consequently, this study proposes an LPV $\mathcal{H}_\infty$ synthesis approach for a quarter-car active suspension model. By extending the robust $\mathcal{H}_\infty$ framework into an LPV structure [8], this work systematically balances ride comfort against road holding across varying operational states, benchmarking the proposed performance against standard passive and LQR-augmented systems.

2 Literature Review

The control of automotive suspension systems has been extensively studied over the past several decades, driven by the inherent conflict between isolating passengers from road vibrations and maintaining firm tire-to-road contact. Hrovat [5] provides a comprehensive survey of these developments, highlighting how optimal control theory has been applied to overcome the physical limitations of passive suspensions.

Early and foundational approaches to active and semi-active suspension control heavily favored the Linear Quadratic Regulator (LQR) and related optimal control strategies. Rajamani [4] outlines the fundamental vehicle dynamics and the application of state-feedback LQR controllers to quarter-car and half-car models. Similarly, Butsuen [2] established foundational performance limits and design methodologies for semi-active suspensions. To ensure these controllers perform accurately in simulation, researchers have also deeply investigated the physical modeling of the quarter-car itself, such as Levitt and Zorka’s work [1] on the often-neglected influence of tire damping.

While LQR provides a mathematically optimal solution for known, linear systems, it lacks inherent robustness to unmodeled dynamics and parameter variations. Consequently, robust $\mathcal{H}_\infty$ synthesis emerged as a highly promising alternative. Yamashita et al. [9] demonstrated the successful application of $\mathcal{H}_\infty$ control to active suspensions, proving its ability to strictly bound the worst-case energy amplification of road disturbances. This approach was further refined by Sun et al. [6], who utilized finite-frequency $\mathcal{H}_\infty$ control to specifically target the 4–8 Hz frequency band where passenger discomfort is most acute. Furthermore, as research expanded from quarter-car to full-car models, $\mathcal{H}_\infty$ synthesis proved vital for managing complex, coupled multi-variable problems, such as integrating active braking to improve vehicle roll stability [3].

Despite the robustness of standard $\mathcal{H}_\infty$ controllers, they are typically designed around a single nominal operating point. Vehicle dynamics, however, are highly non-linear and subject to varying operational states (e.g., changing road roughness and suspension stroke limits). To bridge this gap, Linear Parameter Varying (LPV) control was introduced. Following the foundational theoretical framework for self-scheduled $\mathcal{H}_\infty$ control of LPV systems laid out by Apkarian et al. [10], LPV techniques were rapidly adapted for vehicle suspensions. Fialho and Balas [7] successfully developed a road-adaptive active suspension using LPV gain-scheduling, utilizing suspension deflection as a scheduling parameter. Similarly, Poussot-Vassal et al. [8] demonstrated how LPV techniques can be leveraged to effectively manage the nonlinear constraints of semi-active actuators.

Building upon this rich progression of control methodologies, this study utilizes the LPV $\mathcal{H}_\infty$ framework to dynamically schedule a robust controller, ensuring optimal trade-offs between ride comfort and road holding across a diverse spectrum of ISO 8608 road profiles.

3 Work

3.1 Quarter-car Model

Quarter-car model is an abstraction of an automobile which consist of tire (generally only as stiffness), the car chassis, a damper, spring and actuator between them. It is widely used for benchmarking and design studies of active suspension systems. In Table 1 used parameters can be found.

Table 1: Nominal ('nom') Quarter-Car Model Parameters

Parameter	Symbol	Value
Sprung mass	m_s	240
Unsprung mass	m_u	36
Suspension damping	b_s	1000
Suspension stiffness	k_s	16000
Tire stiffness	k_t	160000
Tire damping	b_t	0

The quarter-car system dynamics can be represented in state-space form as:

$$\dot{x} = Ax + B_x u \quad (1)$$

$$y = Cx + Du \quad (2)$$

Where the state vector x , input vector u , and output vector y are defined as:

$$x = \begin{bmatrix} \text{rattle space} \\ \text{body velocity} \\ \text{tire deflection} \\ \text{tire velocity} \end{bmatrix}, \quad u = \begin{bmatrix} f_s \\ \dot{r} \end{bmatrix}, \quad y = \begin{bmatrix} \text{body acceleration} \\ \text{rattle space} \\ \text{tire deflection} \end{bmatrix} \quad (3)$$

The system matrices are given in (4), (5), (6), (7) ([4], [2]).

$$A = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -\frac{k_s}{m_s} & -\frac{b_s}{m_s} & 0 & \frac{b_s}{m_s} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_u} & \frac{b_s}{m_u} & -\frac{k_t}{m_u} & -\frac{b_s+b_t}{m_u} \end{bmatrix} \quad (4)$$

$$B_x = [B \quad L] = \begin{bmatrix} 0 & 0 \\ \frac{1}{m_s} & 0 \\ 0 & -1 \\ -\frac{1}{m_u} & 0 \end{bmatrix} \quad (5)$$

$$C = \begin{bmatrix} -\frac{k_s}{m_s} & -\frac{b_s}{m_s} & 0 & \frac{b_s}{m_s} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (6)$$

$$D = \begin{bmatrix} \frac{1}{m_s} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (7)$$

Note that road disturbance velocity, $\dot{r}$, is used as input. This representation will be used for LQR design and corresponding comparisons. Nevertheless, for $\mathcal{H}_\infty$ synthesis using road disturbance, r , is better. Formulation is given in (8), (9), (10), (11), (12). Note that s_d represents suspension deflection and t_d represents tire deflection.

$$x = \begin{bmatrix} \text{body travel (m)} \\ \text{body velocity (m/s)} \\ \text{wheel travel (m)} \\ \text{wheel velocity (m/s)} \end{bmatrix}, \quad u = \begin{bmatrix} r \\ f_s \end{bmatrix}, \quad y = \begin{bmatrix} x_b \\ s_d \\ a_b \\ t_d \end{bmatrix} \quad (8)$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_s}{m_s} & -\frac{b_s}{m_s} & \frac{k_s}{m_s} & \frac{b_s}{m_s} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_u} & \frac{b_s}{m_u} & -\frac{k_s+k_t}{m_u} & -\frac{b_s}{m_u} \end{bmatrix} \quad (9)$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1000}{m_s} \\ 0 & 0 \\ \frac{k_t}{m_u} & -\frac{1000}{m_u} \end{bmatrix} \quad (10)$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ -\frac{k_s}{m_s} & -\frac{b_s}{m_s} & \frac{k_s}{m_s} & \frac{b_s}{m_s} \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (11)$$

$$D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & \frac{1000}{m_s} \\ -1 & 0 \end{bmatrix} \quad (12)$$

3.2 Road Profile

Road profiles are generated using ISO8608 standard and given code, which is in attachments (*roadprofile_fun.m* and *roadpr_ISO8608.m*, note that only change to the given code is that it is turned into a function for ease of use). It is the global standard used to classify how rough or bumpy a road is by representing physical bumps as power spectral density (PSD) (Table 2).

Table 2: Road surface classification and ISO8608 PSD parameters

Road Type	G_{h_0} (m ³ /cycle)	k_{road}
A	1×10^{-6}	2
B	4×10^{-6}	2
C	16×10^{-6}	2
D	64×10^{-6}	2
E	256×10^{-6}	2
F	1024×10^{-6}	2
G	4096×10^{-6}	2
H	16384×10^{-6}	2

3.3 LQR Design

For LQR design, equations and parameters from [2] and [4] are used.

The weighting matrix Q , cross-weighting vector N , and control weighting scalar R are defined as:

$$Q = \begin{bmatrix} \frac{k_s^2}{m_s^2} + \rho_1 & \frac{b_s k_s}{m_s^2} & 0 & -\frac{b_s k_s}{m_s^2} \\ \frac{b_s k_s}{m_s^2} & \frac{b_s^2}{m_s^2} + \rho_2 & 0 & -\frac{b_s^2}{m_s^2} \\ 0 & 0 & \rho_3 & 0 \\ -\frac{b_s k_s}{m_s^2} & -\frac{b_s^2}{m_s^2} & 0 & \frac{b_s^2}{m_s^2} + \rho_4 \end{bmatrix} \quad (13)$$

$$N = \begin{bmatrix} -\frac{k_s}{m_s^2} \\ \frac{b_s}{m_s^2} \\ 0 \\ \frac{b_s}{m_s^2} \end{bmatrix}, \quad R = \frac{1}{m_s^2} \quad (14)$$

The LQR gain matrix is computed as:

$$K_{\text{lqr}} = \text{lqr}(A, B, Q, R, N) \quad (15)$$

The closed-loop system matrix is:

$$A_{\text{cl}} = A - BK_{\text{lqr}} \quad (16)$$

The closed-loop output matrix is:

$$C_{\text{cl}} = \begin{bmatrix} A_{\text{cl}}(2, :) \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (17)$$

The closed-loop state-space model is:

$$\text{quartercar_cl_lqr} = \text{ss}(A - BK_{\text{lqr}}, L, C_{\text{cl}}, D(:, 2)) \quad (18)$$

with state names {rattle space, $\dot{x}_b$, tire deflection, $\dot{x}_t$ }, input $\dot{r}$, and outputs $\{\ddot{x}_b$, rattle space, tire deflection, $\ddot{x}_t\}$.

3.4 $\mathcal{H}_\infty$ Synthesis

The actuator is modeled as a first-order low-pass filter with a bandwidth of 60 rad/s. Note that this is a very simple model for an actuator. Normally, real-world actuators inherit complex nonlinearities (such as friction and saturation); nevertheless, for the scope of this research, the actuator model is kept linear, though it remains more physically accurate than a simple DC gain.

$$G_{\text{act}}(s) = \frac{1}{\frac{1}{60}s + 1} \quad (19)$$

The fundamental vehicle frequencies, the body-bounce and wheel-hop natural frequencies, as well as the fixed road filter cut-off frequency (set to 2 Hz, which is chosen considering bounce frequency and importance of high-frequency disturbances) are defined as:

$$\omega_{\text{bounce}} = \sqrt{\frac{k_s}{m_s}}, \quad \omega_{\text{wheel-hop}} = \sqrt{\frac{k_t}{m_u}}, \quad \omega_{\text{road-cut}} = 4\pi \quad (20)$$

The road disturbance is shaped by a first-order filter driven by unit white noise, utilizing the velocity-dependent PSD magnitude:

$$W_{\text{road}}(s) = \frac{\sqrt{G_{h_0} V} \omega_{\text{road_cut}}}{s + \omega_{\text{road_cut}}} \quad (21)$$

The performance weighting functions are explicitly chosen to shape the closed-loop frequency response. The control effort weight penalizes high-frequency actuation to prevent actuator saturation, rolling off above the body-bounce frequency:

$$W_{\text{act}}(s) = 0.1 \frac{s + \omega_{\text{bounce}}}{s + 100} \quad (22)$$

Sensor noise weights on the rattle-space and body-acceleration measurements are modeled as small constants:

$$W_{d_2} = 0.01, \quad W_{d_3} = 0.01 \quad (23)$$

To prevent the suspension from hitting its mechanical limits, the rattle-space (suspension deflection) weight is defined as a low-pass filter scaled by the inverse of the maximum allowable physical travel ($\text{max_travel} = 1.0 \text{ m}$):

$$W_{\text{sd}}(s) = \frac{1}{\text{max_travel}} \cdot \frac{1}{\frac{1}{\omega_{\text{bounce}}} s + 1} \quad (24)$$

Ride comfort is overwhelmingly dictated by body acceleration, but human sensitivity to vertical vibration is highly frequency-dependent, peaking in the 4–8 Hz (approx. 25.1–50.3 rad/s) range. Therefore, the body-acceleration weight W_{ab} is designed as a strict band-pass filter centered at $\omega_{\text{center}} = 37.7 \text{ rad/s}$ with a bandwidth $\Delta\omega_{\text{ab}} = 25.2 \text{ rad/s}$. This heavily amplifies and penalizes accelerations specifically within the human-sensitive bandwidth (as will be seen in Figure 1. It is scaled by $\kappa(1 - \beta)$ to prioritize comfort when $\beta \rightarrow 0$:

$$W_{\text{ab}}(s) = \kappa(1 - \beta) \cdot \frac{\Delta\omega_{\text{ab}} s}{s^2 + \Delta\omega_{\text{ab}} s + \omega_{\text{center}}^2} \quad (25)$$

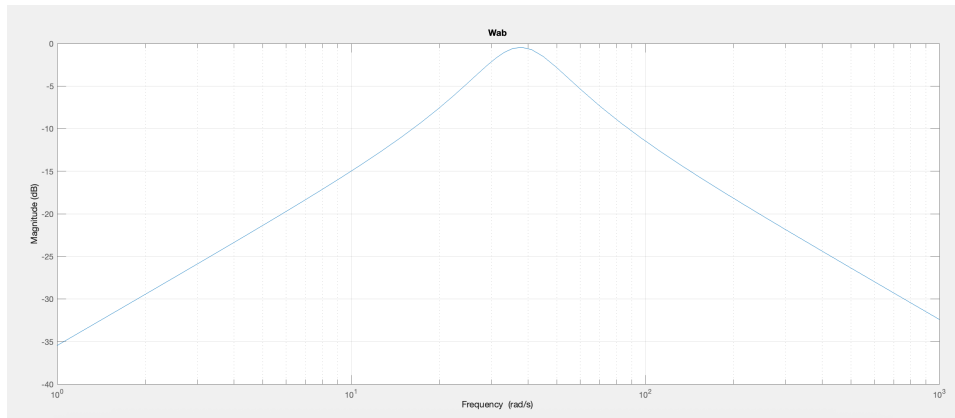


Figure 1: Bode plot of Wab filter.

Conversely, road handling requires maintaining continuous tire contact with the ground. This phenomenon is dominated by the unsprung mass dynamics. Thus, the tire deflection

weight W_{td} is formulated to penalize variations near the wheel-hop frequency. It is scaled by $\kappa\beta$ to aggressively enforce road holding when $\beta \rightarrow 1$:

$$W_{td}(s) = \kappa\beta \cdot \frac{s+1}{s + \omega_{\text{wheel-hop}}} \quad (26)$$

where κ is a global tuning knob for synthesis and $\beta \in [0, 1]$ dynamically trades off ride comfort against road-holding.

The interconnected generalized plant P is formed by connecting the open-loop quarter-car model, the actuator (19), the road filter (21), and all performance weights (22)–(26), with measurement outputs:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_{sd} + d_2 \\ \ddot{x}_b + d_3 \end{bmatrix} \quad (27)$$

and exogenous inputs $\mathbf{w} = [d_1, d_2, d_3]^\top$ alongside the control input u .

The $\mathcal{H}_\infty$ optimal controller is obtained by solving:

$$\min_{K \text{ stabilising}} \|F_l(P, K)\|_\infty = \gamma^* \quad (28)$$

yielding a two-input (y_1, y_2), one-output (u) controller $K(s)$ with achieved $\mathcal{H}_\infty$ norm γ^* .

Reconnecting $K(s)$ with the full plant gives the closed-loop transfer matrix from road input $\dot{r}$ to the four performance outputs:

$$\mathbf{z}_{cl} = \begin{bmatrix} \ddot{x}_b \\ x_{sd} \\ x_b \\ x_{td} \end{bmatrix} = T_{cl}(s) r \quad (29)$$

3.5 Methodology

To rigorously evaluate the proposed control strategies, a systematic simulation framework was established. The quarter-car model, along with the nominal parameters derived from Rajamani [4] and Butsuen [2], serves as the foundational plant. The unaugmented open-loop (passive) system and an LQR-controlled system were utilized as baseline benchmarks, representing standard industry approaches.

The evaluation protocol is divided into qualitative frequency-domain analysis and quantitative time-domain simulations:

- **Qualitative Evaluation (Frequency Domain):** Bode plots are utilized to analyze the closed-loop transfer functions. The primary objective is to verify that the control effort successfully suppresses gains within critical frequency bands, specifically the 4–8 Hz range (which correlates to human sensitivity and passenger comfort) and the wheel-hop frequency (which dictates road holding).
- **Quantitative Evaluation (Time Domain):** The vehicle is simulated traveling across standardized ISO 8608 road profiles ranging from Class A (very smooth) to Class H (extremely rough). The primary performance metrics extracted from these simulations are the Root Mean Square (RMS) values of the body acceleration (a_b) and tire deflection (t_d), serving as proxies for ride comfort and road holding, respectively.

A critical safety and stability constraint enforced throughout the methodology is the prevention of tire lift-off. To ensure continuous road contact, the dynamic tire force must not exceed the static load of the vehicle. Speeds for each road profile were bounded using a statistical 3σ criterion ($3\sigma_{F_{\text{dyn}}} < F_{\text{static}}$), ensuring the probability of lift-off remains strictly negligible (below 0.13%).

Parametric Sweep and Cross-Validation: The core of the $\mathcal{H}_\infty$ synthesis involves a global tuning parameter, $\beta \in [0, 1]$, which dictates the strict trade-off between ride comfort ($\beta \rightarrow 0$) and road holding ($\beta \rightarrow 1$). To thoroughly assess controller robustness, an iterative parameter sweep was conducted across β values (e.g., 0.05, 0.50, 0.95). Furthermore, a comprehensive cross-simulation testing matrix was developed. Controllers synthesized specifically for one road class (e.g., K_A) were evaluated against all other road classes (A through H). This cross-validation determines how well a static $\mathcal{H}_\infty$ controller performs when subjected to off-design road disturbances.

Finally, based on the limits of static robust control identified in the cross-validation, an interpolation-based Linear Parameter Varying (LPV) gain-scheduling approach is implemented. This allows the system to dynamically shift the β parameter in real-time, optimizing the trade-off based on instantaneous road conditions.

4 Results

4.1 Baseline Analysis

The evaluation begins by establishing a baseline using the open-loop (passive) system and a standard LQR-controlled system. Figure 2 illustrates the Bode magnitude plots for these baseline configurations. The LQR approach successfully dampens the primary body-bounce resonance compared to the passive system, establishing a standard benchmark for evaluating further ride comfort improvements.

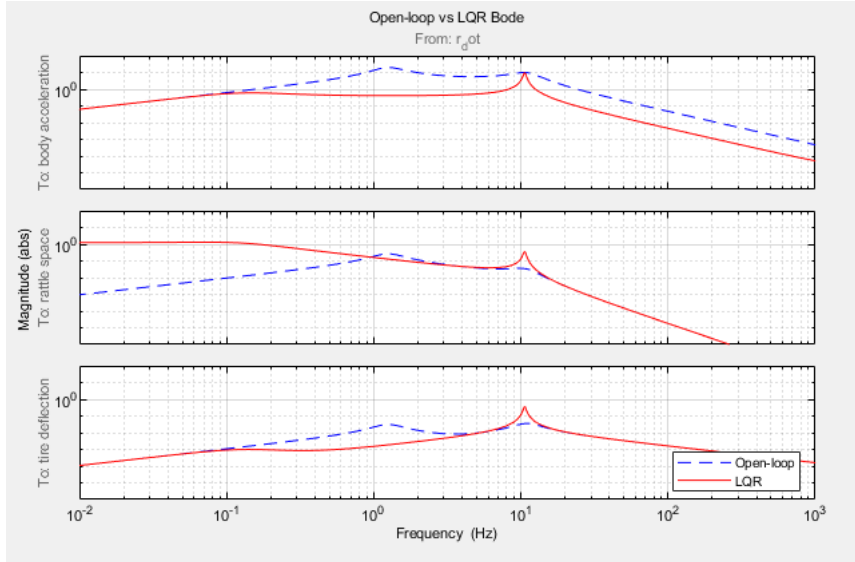


Figure 2: Bode plot comparison between open-loop and LQR augmented systems.

4.2 Road Profile - Speed Matching

Before applying robust control techniques, realistic operational boundaries must be established. The driving speed for each road class was derived from a physical road-holding limit rather than assigned arbitrarily, since a vehicle must instinctively slow down as the surface worsens to maintain safety. For each ISO 8608 class, the passive quarter-car was simulated over the generated profile and the dynamic tire force $F_{\text{dyn}}(t) = k_t \delta_t(t)$ was evaluated, with lift-off occurring when it cancels the static load $F_{\text{static}} = (m_s + m_u)g$.

To avoid keying the limit to a single peak of the random profile, a statistical 3σ criterion was enforced as a primary design consideration: a speed is deemed admissible while $3\sigma_{F_{\text{dyn}}} < F_{\text{static}}$. Under an approximately Gaussian force assumption, this keeps the theoretical lift-off probability strictly below 0.13%. The maximum admissible speed was found by bisection on speed with the random profile held fixed via a constant seed. Speeds for the two smoothest classes (A, B) were capped at a practical highway search bound of 60 m/s, as no lift-off occurred. The calculated speed limits and corresponding road roughness values are detailed in Table 3.

Table 3: Maximum admissible speeds and road roughness limits determined per ISO 8608 class.

Road Type	Road Roughness G_{h_0} [m^3/rad]	Speed [m/s]
Road A	1×10^{-6}	60.00
Road B	4×10^{-6}	60.00
Road C	16×10^{-6}	33.36
Road D	64×10^{-6}	9.00
Road E	256×10^{-6}	4.10
Road F	1024×10^{-6}	2.97
Road G	4096×10^{-6}	1.46
Road H	16384×10^{-6}	0.57

4.3 Preliminary $\mathcal{H}_\infty$ Synthesis

With physical constraints defined, the $\mathcal{H}_\infty$ synthesis was initially applied using the Road Type A profile as a nominal average-case design. As observed in the frequency response in Figure 3, the $\mathcal{H}_\infty$ controller actively shapes the system's output, aggressively suppressing gain magnitudes across the critical passenger comfort frequency band while simultaneously respecting the tire lift-off constraints.

4.4 Time-Domain Comparisons

To validate these frequency-domain characteristics, a comparative time-domain simulation was executed using the significantly rougher Road Type D profile. Figure 4 demonstrates the transient responses of the open-loop, LQR, and $\mathcal{H}_\infty$ systems. The robust $\mathcal{H}_\infty$ architecture demonstrates good peak attenuation and faster disturbance rejection when subjected to severe road irregularities.

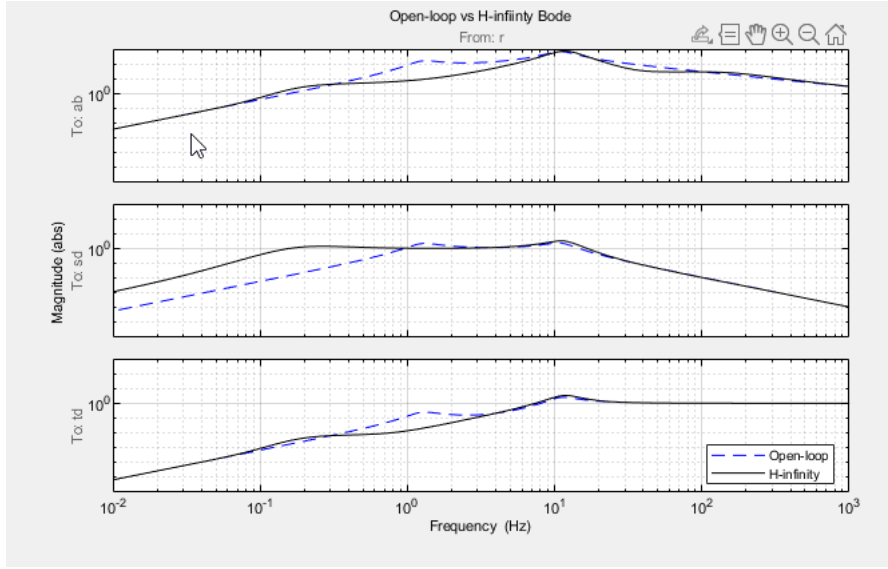


Figure 3: Bode plot evaluation for the $\mathcal{H}_\infty$ controller synthesized for Road Type A.

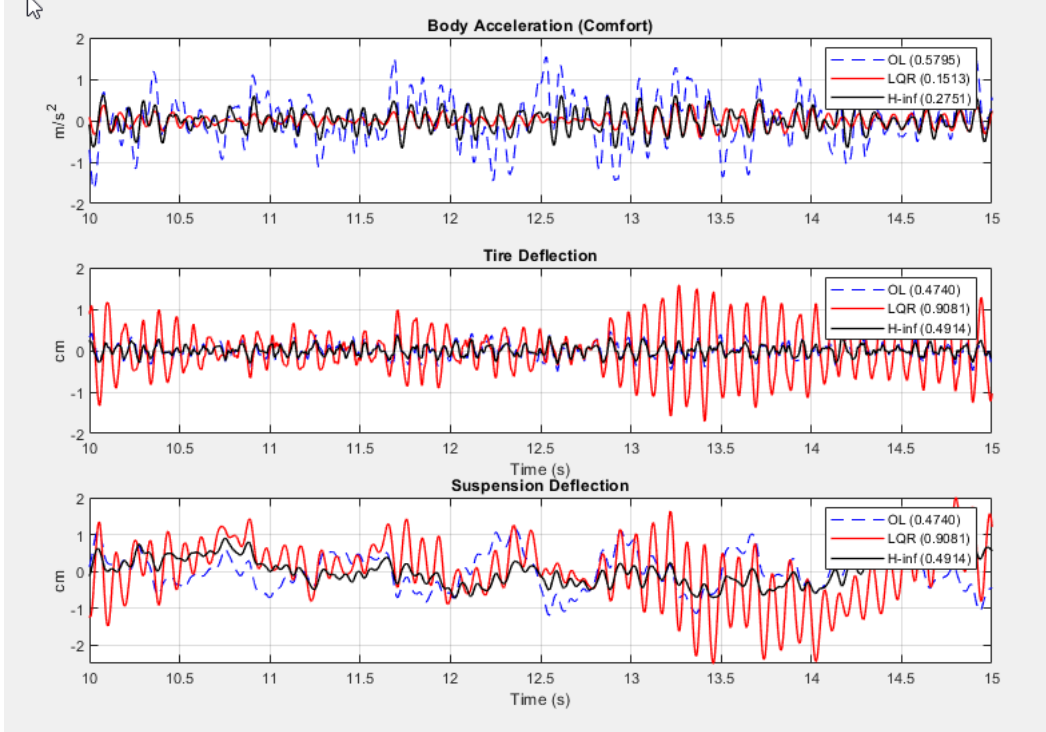


Figure 4: Time-domain simulation comparison of open-loop, LQR, and $\mathcal{H}_\infty$ strategies evaluated on Road Type D.

4.5 β -Sweep

To systematically explore the performance boundaries and flexibility of the $\mathcal{H}_\infty$ framework, an iterative parametric sweep was conducted over the performance tuning weight $\beta \in [0, 1]$. This parameter mathematically dictates the trade-off between passenger comfort and vehicle handling. The sweep was repeated across all road classes (A through H), with the visual frequency responses captured in Figures 5 through 12.

Quantitatively, the resulting Root Mean Square (RMS) values for body acceleration (a_b) and tire deflection (t_d) are summarized in Table 4. The data clearly follows the theoretical expectation: lower values of β (e.g., 0.05) strictly prioritize passenger comfort, yielding the lowest a_b RMS values but allowing greater tire deflection. Conversely, higher values (e.g., 0.95) strictly enforce road holding, minimizing the t_d RMS at the expense of a harsher ride. Note that the relation where as β increases road handling gets better is lost after Road F, this should be due to fact that those profiles are really rough and relation between comfort and handling gets nonlinear, so this case should be investigated further.

Another important thing is that on all scenarios, even where β is 0.95, tire deflection RMS is worse than open-loop's. This may be due to a poor filter design.

Table 4: RMS performance of the $\mathcal{H}_\infty$ β -sweep controllers (a_b : body acceleration [m/s²], t_d : tire deflection [cm])

β	Road A		Road B	
	a_b	t_d	a_b	t_d
OL	0.6046	0.1918	1.2092	0.3836
0.05	0.2653	0.4507	0.7521	0.6897
0.27	0.2707	0.4441	0.7639	0.6817
0.50	0.2781	0.4353	0.7743	0.6751
0.72	0.3113	0.4012	0.8070	0.6561
0.95	0.9518	0.2197	1.7684	0.4395

β	Road C		Road D	
	a_b	t_d	a_b	t_d
OL	1.8090	0.5641	1.9005	0.5640
0.05	1.2100	0.9674	1.5125	0.8976
0.27	1.3903	0.8940	1.5241	0.8915
0.50	1.4119	0.8823	1.4960	0.9077
0.72	1.4666	0.8572	1.5548	0.8810
0.95	2.6817	0.6678	2.7304	0.6794

β	Road E		Road F	
	a_b	t_d	a_b	t_d
OL	2.3250	0.5635	3.2076	0.5635
0.05	1.7733	1.0128	2.7997	0.5577
0.27	1.7406	1.0388	2.7922	0.5574
0.50	1.7536	1.0290	2.7721	0.5564
0.72	1.7783	1.0138	2.8650	0.5613
0.95	3.0404	0.6939	3.5506	0.6189

β	Road G		Road H	
	a_b	t_d	a_b	t_d
OL	3.7437	0.5635	3.6162	0.5657
0.05	4.2134	0.6012	4.8311	0.7266
0.27	4.1778	0.5961	4.8110	0.7231
0.50	4.1877	0.5975	4.8215	0.7249
0.72	4.1788	0.5962	4.8291	0.7260
0.95	4.7973	0.6880	5.2940	0.7966

4.6 Cross Analysis

While a dedicated $\mathcal{H}_\infty$ controller performs optimally on its specifically designed road profile, practical vehicles encounter dynamically changing road types. To rigorously evaluate robust performance boundaries and set the motivation for adaptive control (LPV), a comprehensive cross-simulation testing matrix was executed. In this setup, controllers synthesized for a specific road condition (K_A through K_H) were subjected to all possible off-design road profiles. This cross-validation was tested across three distinct tuning extremes: comfort-focused ($\beta = 0.05$), balanced ($\beta = 0.50$), and handling-focused ($\beta = 0.95$).

The visual time-domain distributions for these cross-tests are documented in Figures 13 to 36. Even though K_A seems to be winning on all conditions, when tables for rattle space checked, it can be seen that comfort is achieved for the sake of unrealistic efforts. For this reason, a triple-tuning scheme (acceleration, road handling and suspension deflection) should be done for more realistic results.

Table 5: Cross-test body-acceleration RMS [m/s²], $\beta = 0.05$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.605	1.209	1.809	1.901	2.325	3.208	3.744	3.616
K_A	0.265	0.531	0.923	0.991	1.294	1.740	2.412	2.976
K_B	0.376	0.752	1.228	1.303	1.649	2.267	3.122	3.816
K_C	0.388	0.775	1.210	1.284	1.606	2.135	2.929	3.538
K_D	0.454	0.907	1.426	1.512	1.874	2.618	3.593	4.339
K_E	0.442	0.884	1.352	1.439	1.773	2.400	3.285	3.915
K_F	0.519	1.038	1.555	1.665	2.015	2.800	3.820	4.426
K_G	0.569	1.138	1.694	1.818	2.189	3.098	4.213	4.731
K_H	0.590	1.180	1.755	1.883	2.267	3.233	4.388	4.831

Table 6: Cross-test tire-deflection RMS [cm], $\beta = 0.05$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.192	0.384	0.564	0.564	0.564	0.564	0.564	0.566
K_A	0.451	0.901	1.319	1.370	1.728	0.527	0.368	0.447
K_B	0.345	0.690	1.009	1.042	1.239	0.548	0.457	0.566
K_C	0.331	0.662	0.967	0.998	1.171	0.532	0.425	0.525
K_D	0.298	0.595	0.873	0.898	1.019	0.564	0.520	0.644
K_E	0.297	0.594	0.870	0.895	1.013	0.542	0.472	0.581
K_F	0.260	0.521	0.764	0.781	0.837	0.558	0.545	0.660
K_G	0.241	0.483	0.708	0.722	0.744	0.572	0.601	0.709
K_H	0.234	0.468	0.687	0.699	0.709	0.578	0.627	0.727

Table 7: Cross-test suspension-deflection (rattle space) RMS [cm], $\beta = 0.05$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.486	0.972	1.472	1.586	2.107	3.471	4.685	4.882
K_A	1.029	2.059	7.348	12.637	17.868	30.570	43.209	53.734
K_B	1.241	2.483	7.625	13.963	19.819	34.207	48.544	60.700
K_C	1.171	2.342	5.983	10.740	15.506	26.714	38.078	47.698
K_D	1.356	2.712	7.322	13.746	19.964	34.467	49.229	61.728
K_E	1.211	2.422	5.742	10.488	15.603	26.869	38.506	48.329
K_F	1.205	2.411	5.175	9.521	15.139	26.142	37.655	47.472
K_G	1.147	2.293	4.613	8.380	13.932	24.250	34.964	44.260
K_H	1.101	2.201	4.306	7.752	13.245	23.262	33.579	42.650

Table 8: Cross-test body-acceleration RMS [m/s²], $\beta = 0.50$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.605	1.209	1.809	1.901	2.325	3.208	3.744	3.616
K_A	0.278	0.556	0.966	1.037	1.351	1.833	2.539	3.131
K_B	0.387	0.774	1.260	1.336	1.687	2.330	3.208	3.917
K_C	0.448	0.895	1.412	1.497	1.858	2.595	3.562	4.307
K_D	0.447	0.895	1.411	1.496	1.856	2.591	3.556	4.301
K_E	0.436	0.872	1.336	1.421	1.754	2.368	3.242	3.871
K_F	0.514	1.028	1.542	1.650	1.999	2.772	3.783	4.393
K_G	0.566	1.133	1.686	1.809	2.178	3.079	4.188	4.714
K_H	0.588	1.176	1.749	1.876	2.258	3.219	4.369	4.821

Table 9: Cross-test tire-deflection RMS [cm], $\beta = 0.50$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.192	0.384	0.564	0.564	0.564	0.564	0.564	0.566
K_A	0.435	0.871	1.273	1.322	1.657	0.531	0.384	0.469
K_B	0.338	0.675	0.988	1.020	1.206	0.552	0.469	0.581
K_C	0.301	0.602	0.882	0.908	1.035	0.563	0.516	0.639
K_D	0.301	0.602	0.882	0.908	1.035	0.563	0.515	0.638
K_E	0.301	0.601	0.880	0.905	1.029	0.540	0.466	0.574
K_F	0.262	0.525	0.769	0.787	0.846	0.556	0.540	0.655
K_G	0.242	0.485	0.711	0.725	0.749	0.571	0.598	0.706
K_H	0.235	0.469	0.689	0.702	0.713	0.577	0.624	0.725

Table 10: Cross-test suspension-deflection (rattle space) RMS [cm], $\beta = 0.50$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.486	0.972	1.472	1.586	2.107	3.471	4.685	4.882
K_A	1.057	2.114	7.607	13.062	18.465	31.614	44.684	55.577
K_B	1.265	2.530	7.682	14.006	19.876	34.307	48.682	60.875
K_C	1.352	2.704	7.400	13.887	20.113	34.729	49.575	62.149
K_D	1.351	2.702	7.394	13.953	20.271	34.999	49.996	62.691
K_E	1.209	2.417	5.787	10.605	15.781	27.176	38.950	48.889
K_F	1.209	2.417	5.223	9.634	15.324	26.462	38.118	48.061
K_G	1.152	2.305	4.655	8.473	14.091	24.529	35.367	44.775
K_H	1.106	2.212	4.340	7.827	13.378	23.495	33.919	43.085

Table 11: Cross-test body-acceleration RMS [m/s²], $\beta = 0.95$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.605	1.209	1.809	1.901	2.325	3.208	3.744	3.616
K_A	0.952	1.904	3.689	3.957	5.070	8.398	11.652	14.374
K_B	0.884	1.768	3.013	3.164	3.922	6.337	8.741	10.680
K_C	0.829	1.658	2.682	2.821	3.440	5.456	7.505	9.081
K_D	0.812	1.623	2.590	2.730	3.317	5.228	7.187	8.661
K_E	0.768	1.536	2.383	2.528	3.040	4.714	6.468	7.703
K_F	0.628	1.257	1.886	2.021	2.422	3.551	4.848	5.596
K_G	0.631	1.261	1.880	2.018	2.417	3.524	4.797	5.388
K_H	0.635	1.270	1.892	2.030	2.437	3.550	4.820	5.294

Table 12: Cross-test tire-deflection RMS [cm], $\beta = 0.95$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.192	0.384	0.564	0.564	0.564	0.564	0.564	0.566
K_A	0.220	0.439	0.738	0.777	0.895	1.283	1.731	2.151
K_B	0.220	0.440	0.684	0.704	0.765	0.984	1.289	1.596
K_C	0.220	0.441	0.668	0.683	0.722	0.859	1.101	1.356
K_D	0.221	0.442	0.665	0.679	0.713	0.828	1.052	1.293
K_E	0.222	0.444	0.660	0.673	0.694	0.757	0.942	1.150
K_F	0.234	0.468	0.688	0.701	0.718	0.619	0.697	0.836
K_G	0.229	0.458	0.673	0.685	0.690	0.608	0.688	0.808
K_H	0.225	0.451	0.662	0.673	0.671	0.605	0.691	0.797

Table 13: Cross-test suspension-deflection (rattle space) RMS [cm], $\beta = 0.95$ (rows: design road, cols: test road)

$K \setminus \text{Test}$	A	B	C	D	E	F	G	H
OL	0.486	0.972	1.472	1.586	2.107	3.471	4.685	4.882
K_A	3.301	6.603	34.606	67.792	96.361	165.878	235.036	292.634
K_B	2.891	5.782	21.343	42.900	61.147	105.888	150.608	188.535
K_C	2.555	5.109	15.940	32.385	47.143	81.589	116.726	146.486
K_D	2.442	4.884	14.509	29.112	42.473	73.479	105.159	131.993
K_E	2.147	4.294	11.456	22.932	34.424	59.459	85.470	107.462
K_F	1.455	2.911	6.461	12.348	19.759	34.168	49.281	62.193
K_G	1.287	2.574	5.311	9.917	16.584	28.900	41.705	52.845
K_H	1.186	2.372	4.705	8.633	14.818	26.047	37.625	47.823

5 Conclusions and Future Work

In this study, robust $\mathcal{H}_\infty$ synthesis was applied to an active quarter-car suspension system and benchmarked against standard passive and LQR-controlled systems. Through rigorous β -parameter sweeping and cross-simulation matrices across ISO 8608 road profiles, the fundamental trade-off between passenger comfort and road handling was thoroughly analyzed. The results clearly demonstrated that while static $\mathcal{H}_\infty$ controllers excel at their nominal design points, their performance inevitably changes when subjected to off-design road conditions and non-linear relations are observed, strongly highlighting the necessity of deeper analysis and adaptive control strategies.

While the initial objective of this project was to implement a rigorous Linear Parameter Varying (LPV) controller to address these dynamic conditions, time constraints necessitated a practical compromise. Consequently, even though different synthesis are done and cross-tested, a working LPV couldn't be achieved, so it was decided to not adding here.

However, the complexities and challenges encountered during this project have strongly motivated my desire to continue studying subject. To rectify current shortcomings and expand the scope of this research, my future work plans include:

- **Rigorous Polytopic LPV Synthesis:** Proper LPV design for the problem using *hinfgs.m* function of matlab.
- **Improving Filters:** Improving filters so that more realistic and succesfull outcomes achieved (rattle space should also be weighted).
- **Actuator Non-linearities:** Incorporating more realistic, non-linear actuator dynamics (such as friction, saturation, and time-delays) into the generalized plant to bridge the gap between pure simulation and real-world physical constraints.

Ultimately, this project served as a profound learning experience, and completing these future objectives will be the next logical step in our continuous study of advanced vehicle dynamics.

References

- [1] Levitt, J. A., and N. G. Zorka. "The influence of tire damping in quarter car active suspension models." (1991): 134-137.
- [2] Butsuen, T. (1989). The design of semi-active suspensions for automotive vehicles [Massachusetts Institute of Technology]. <http://hdl.handle.net/1721.1/14220>
- [3] Sename, Olivier, Luc Dugard, and Peter Gáspár. "The design of an H/LPV active braking control to improve vehicle roll stability." IFAC-PapersOnLine 52.17 (2019): 54-59.
- [4] Rajamani, Rajesh. Vehicle dynamics and control. Boston, MA: Springer US, 2006.
- [5] Hrovat, Davor. "Survey of advanced suspension developments and related optimal control applications." Automatica 33.10 (1997): 1781-1817.
- [6] Sun, Weichao, Huijun Gao, and Okyay Kaynak. "Finite frequency H_∞ control for vehicle active suspension systems." IEEE Transactions on control systems technology 19.2 (2010): 416-422.
- [7] Fialho, Ian, and Gary J. Balas. "Road adaptive active suspension design using linear parameter-varying gain-scheduling." IEEE transactions on control systems technology 10.1 (2002): 43-54.
- [8] Poussot-Vassal, Charles, et al. "A new semi-active suspension control strategy through LPV technique." Control Engineering Practice 16.12 (2008): 1519-1534.
- [9] Yamashita, Masashi, et al. "Application of H control to active suspension systems." Automatica 30.11 (1994): 1717-1729.
- [10] Apkarian, Pierre, Pascal Gahinet, and Greg Becker. "Self-scheduled H control of linear parameter-varying systems: a design example." Automatica 31.9 (1995): 1251-1261.

6 Appendix

6.1 Figures

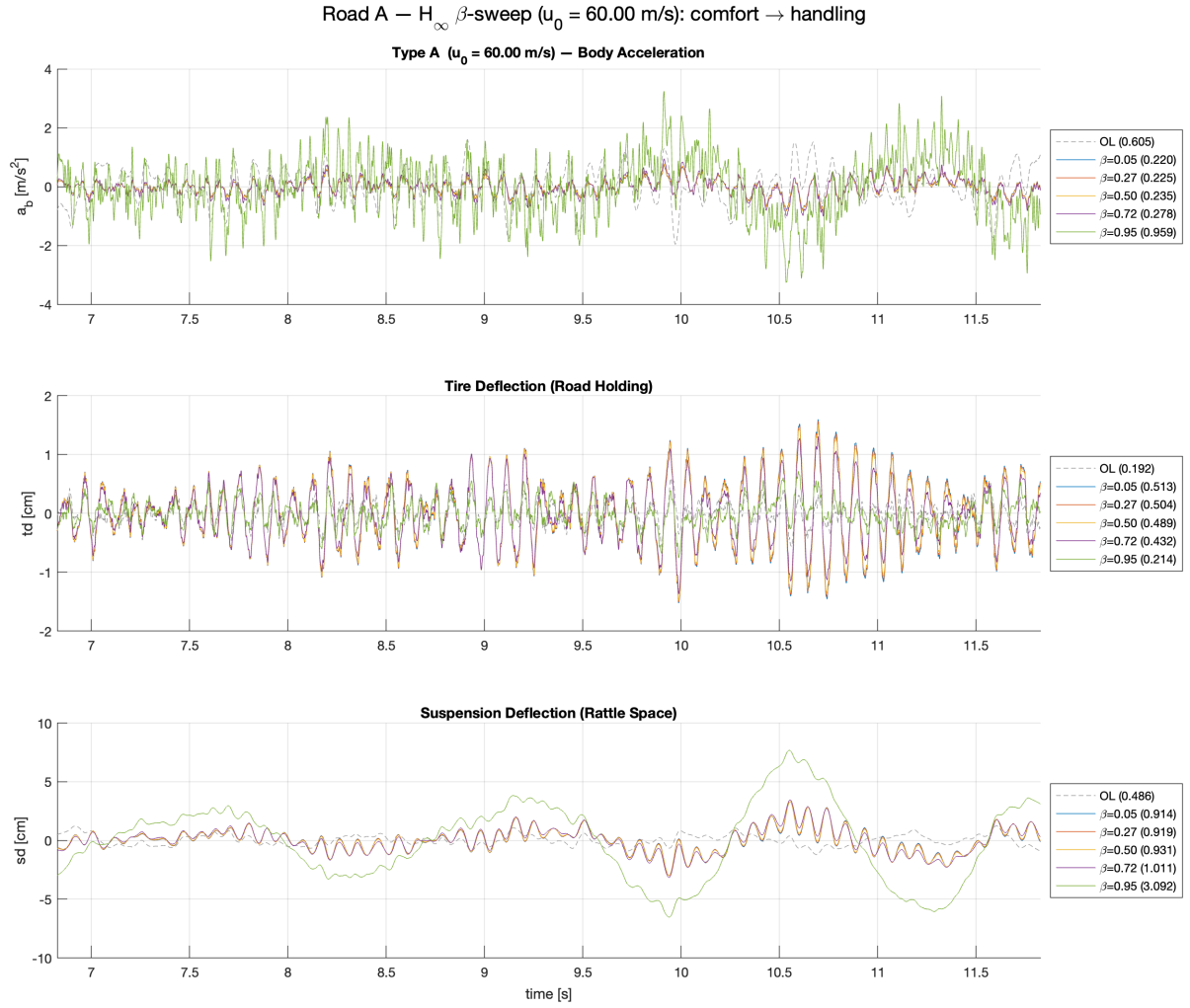


Figure 5: $\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type A.

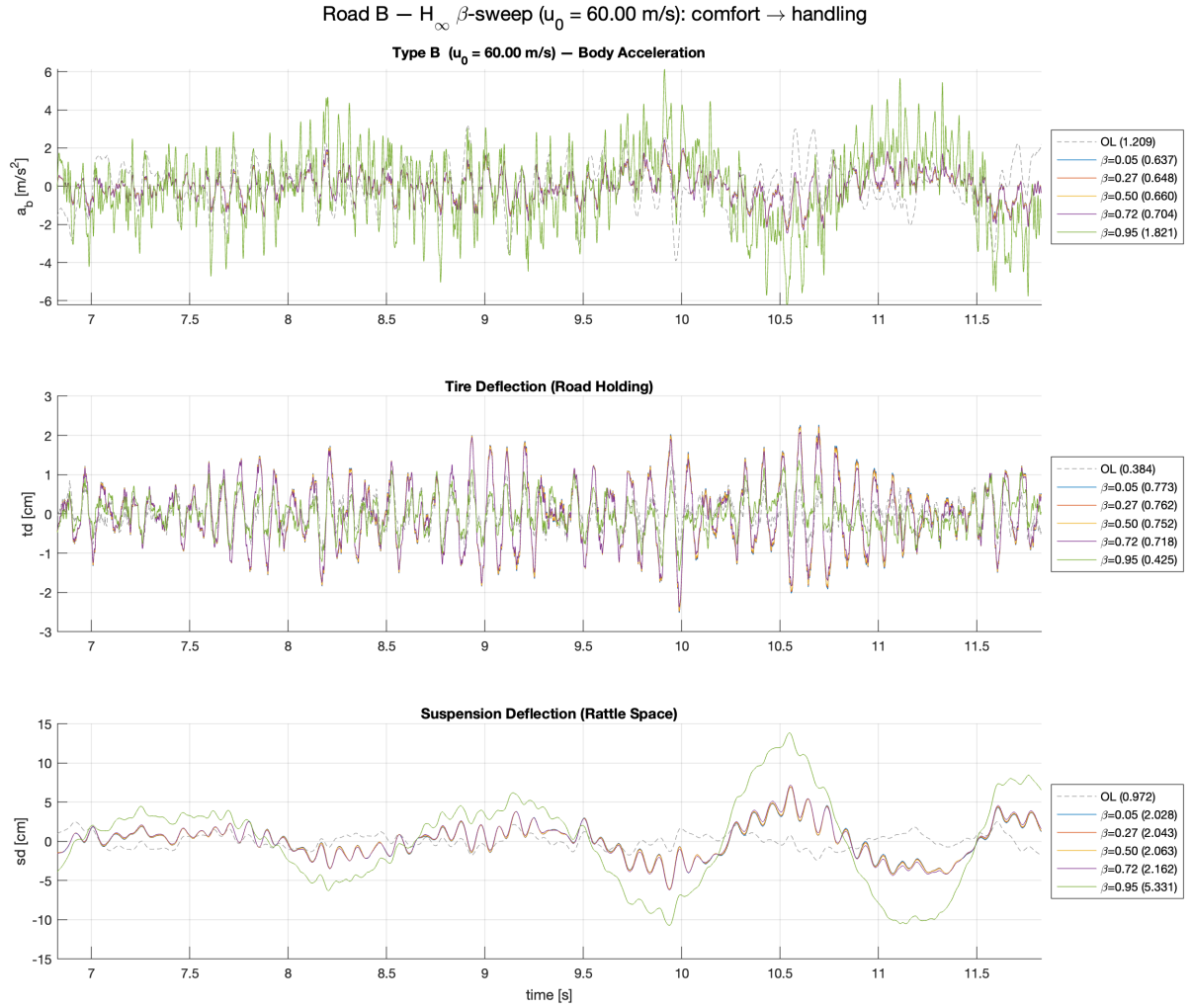


Figure 6: $\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type B.

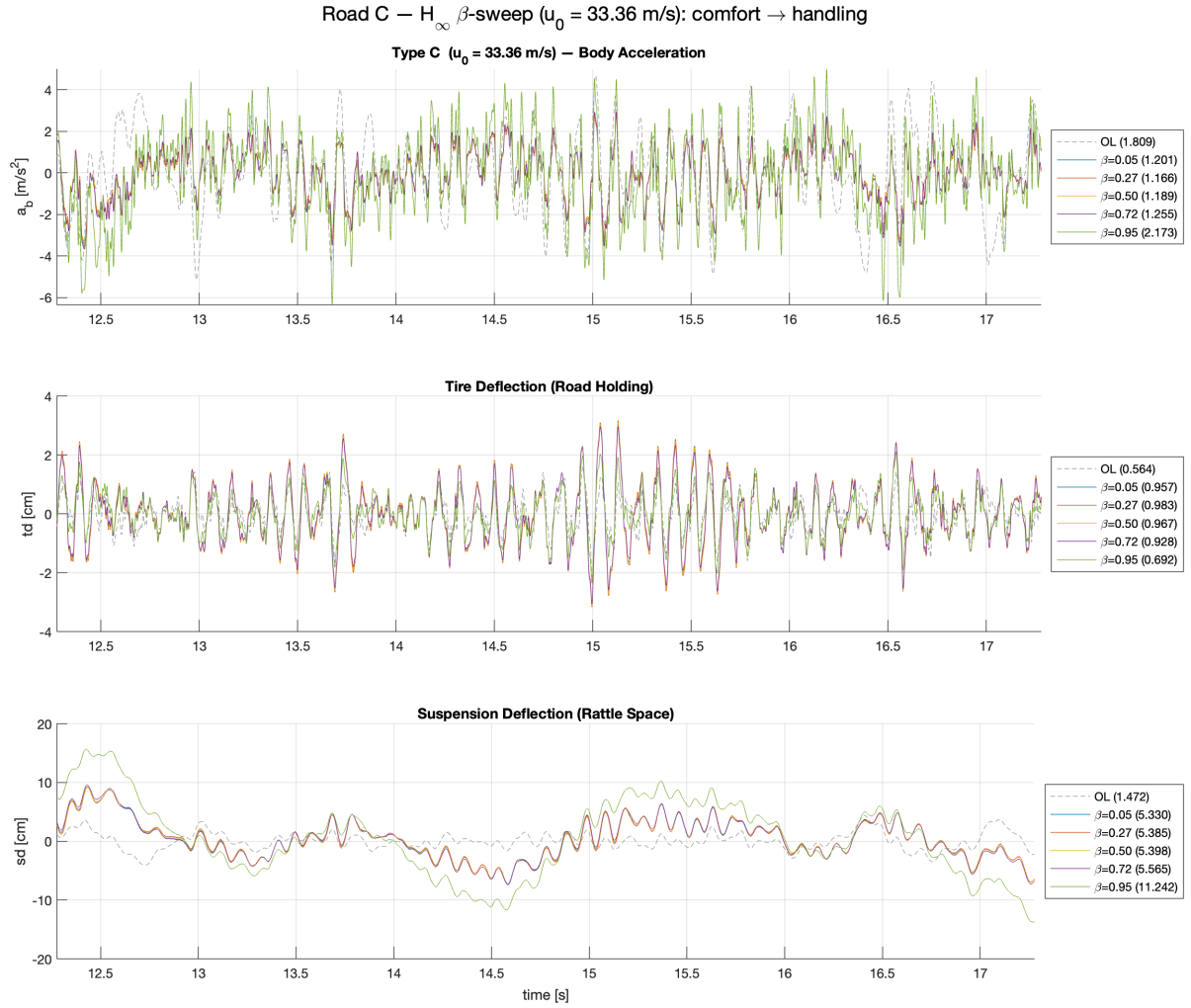


Figure 7: $\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type C.

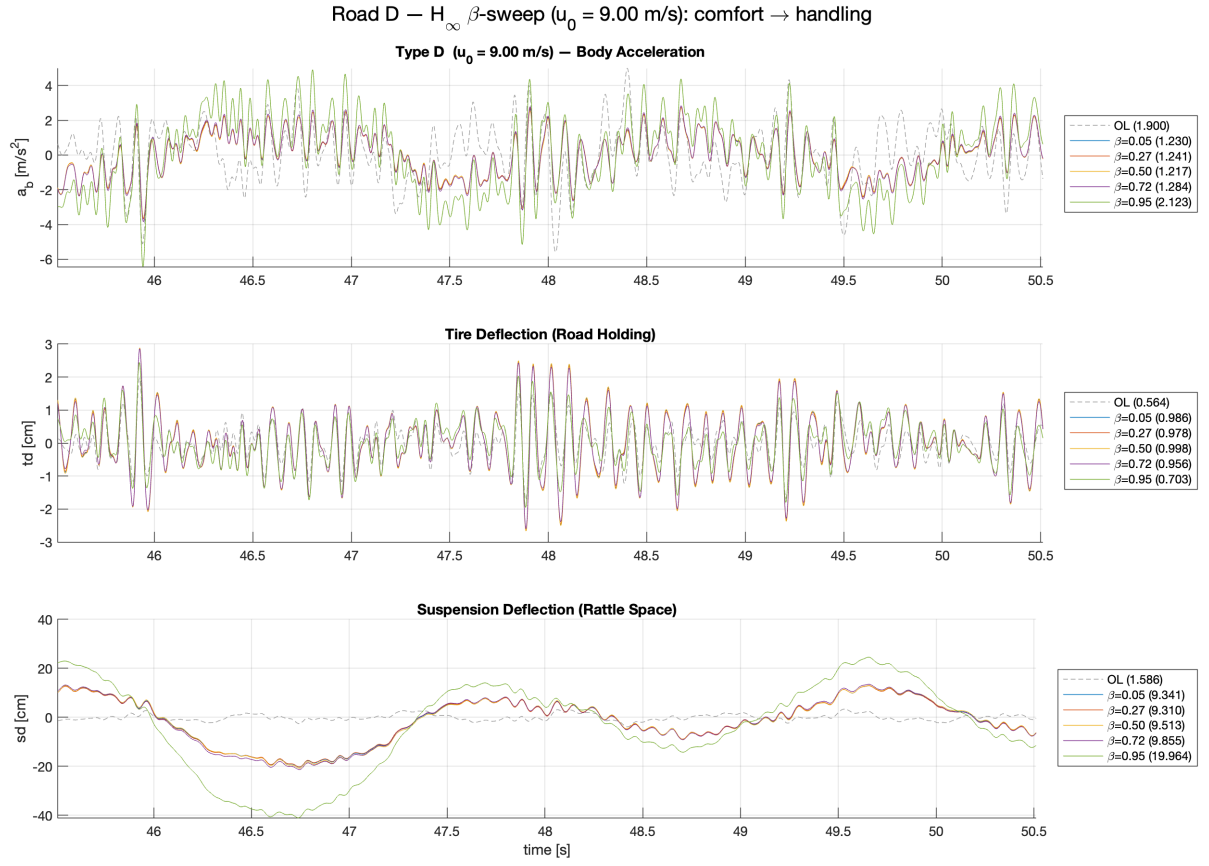


Figure 8: $\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type D.

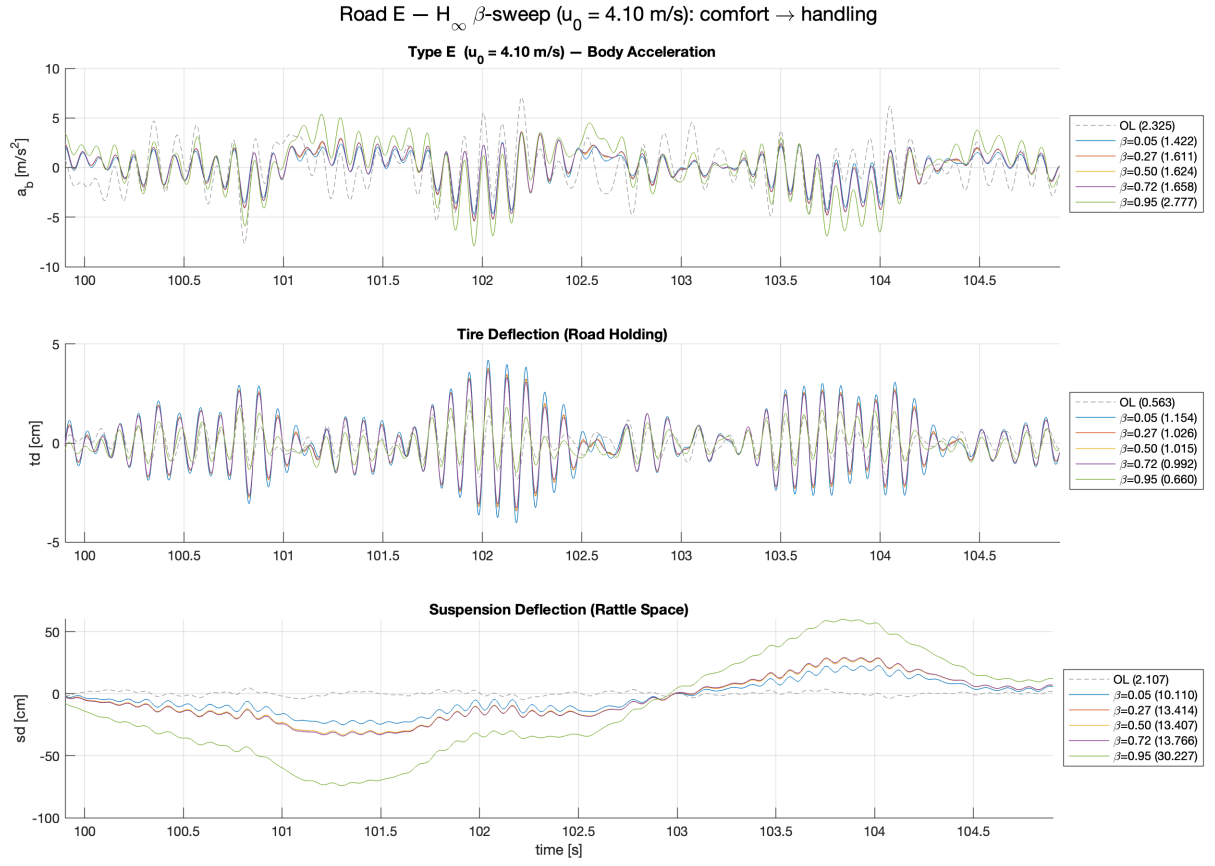


Figure 9: H_∞ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type E.

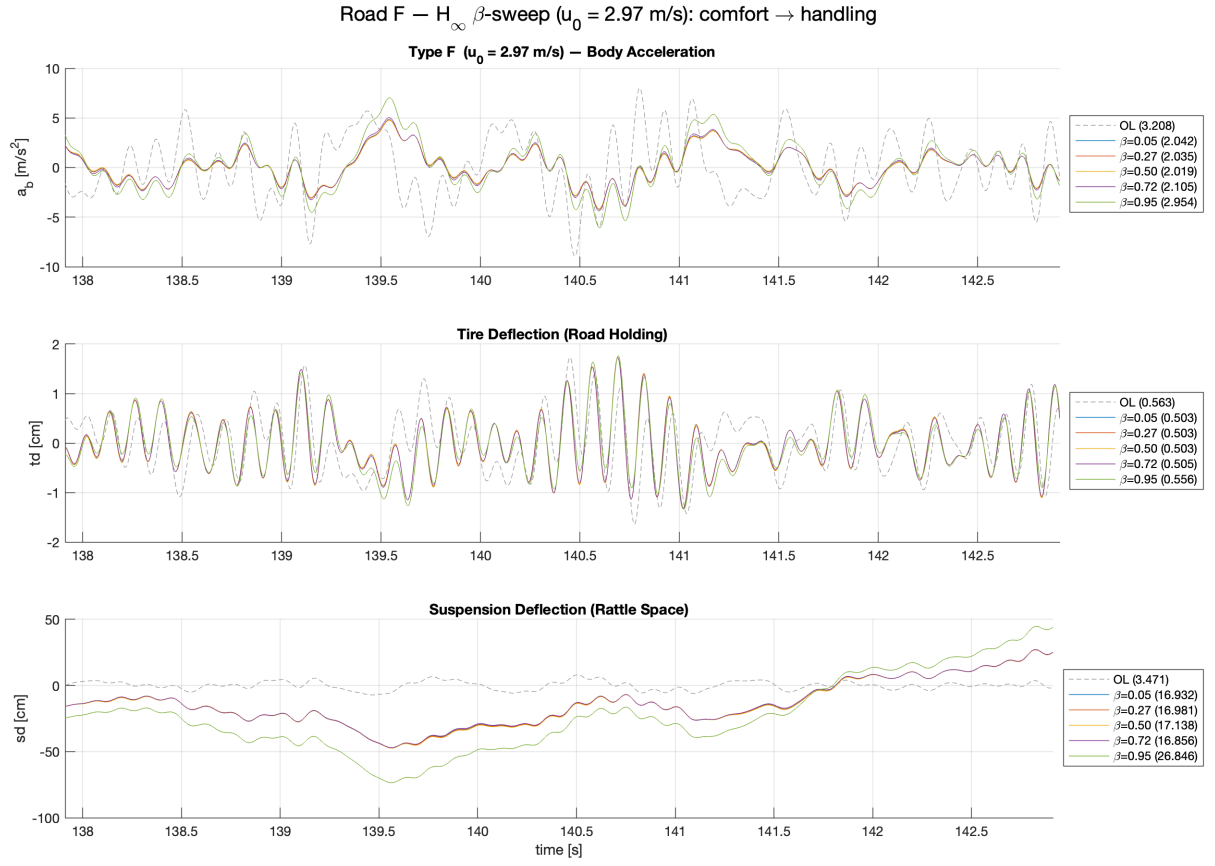


Figure 10: $\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type F.

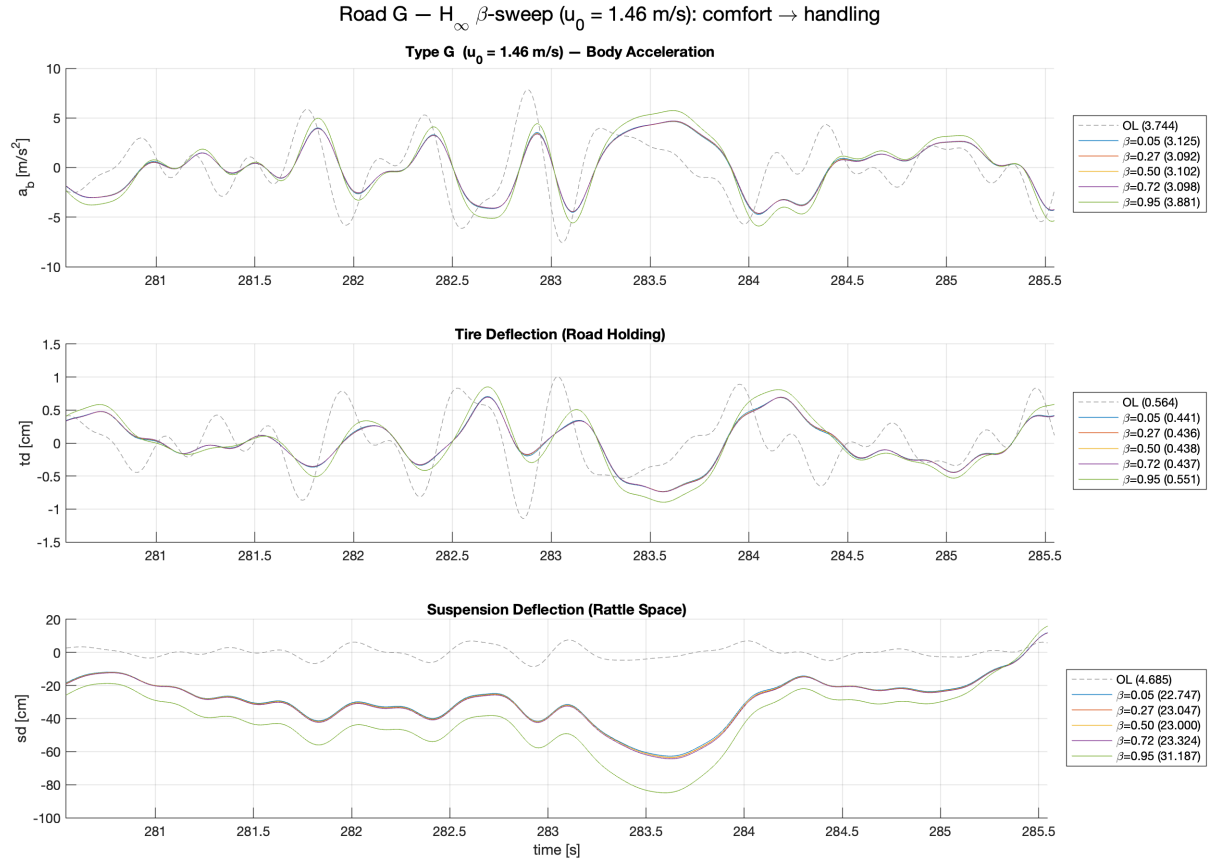


Figure 11: $\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type G.

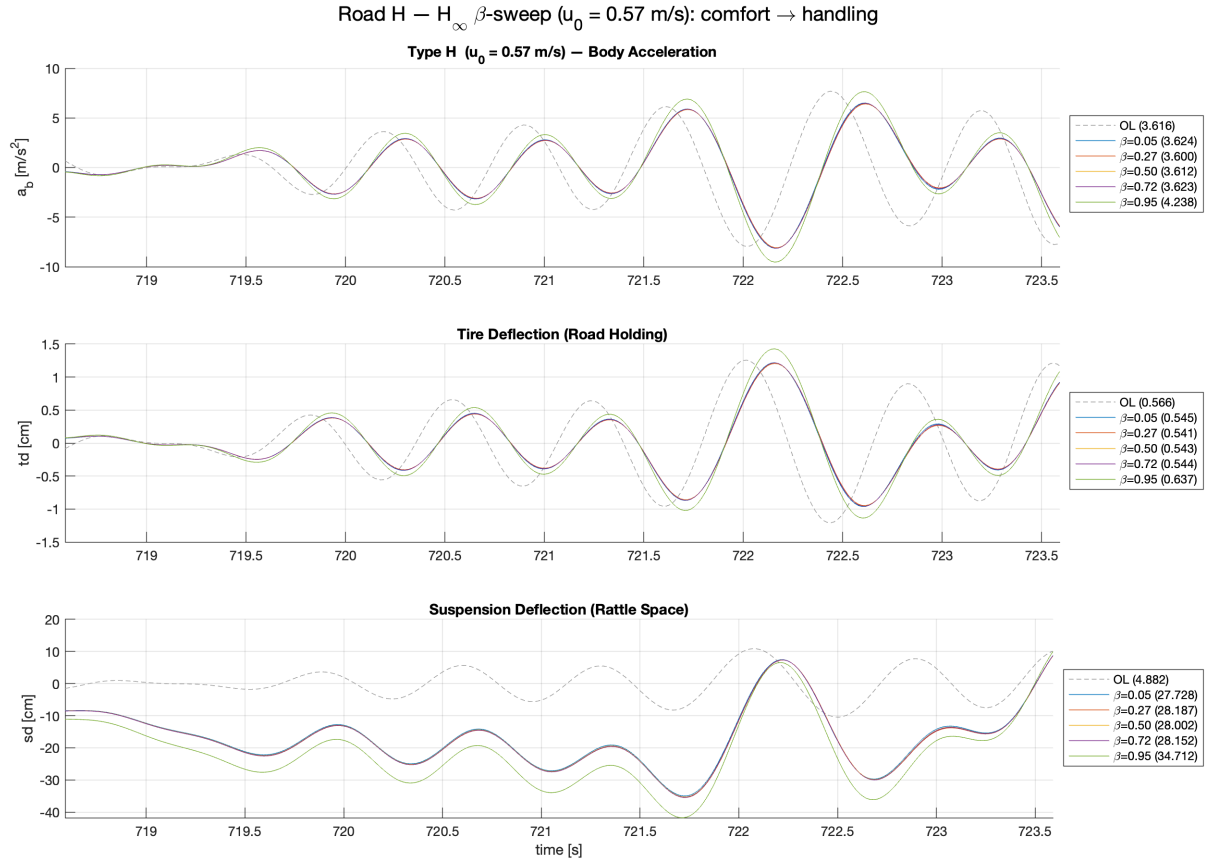


Figure 12: $\mathcal{H}_\infty$ controller β -sweep parameter evaluation (comfort vs. handling trade-off) for Road Type H.

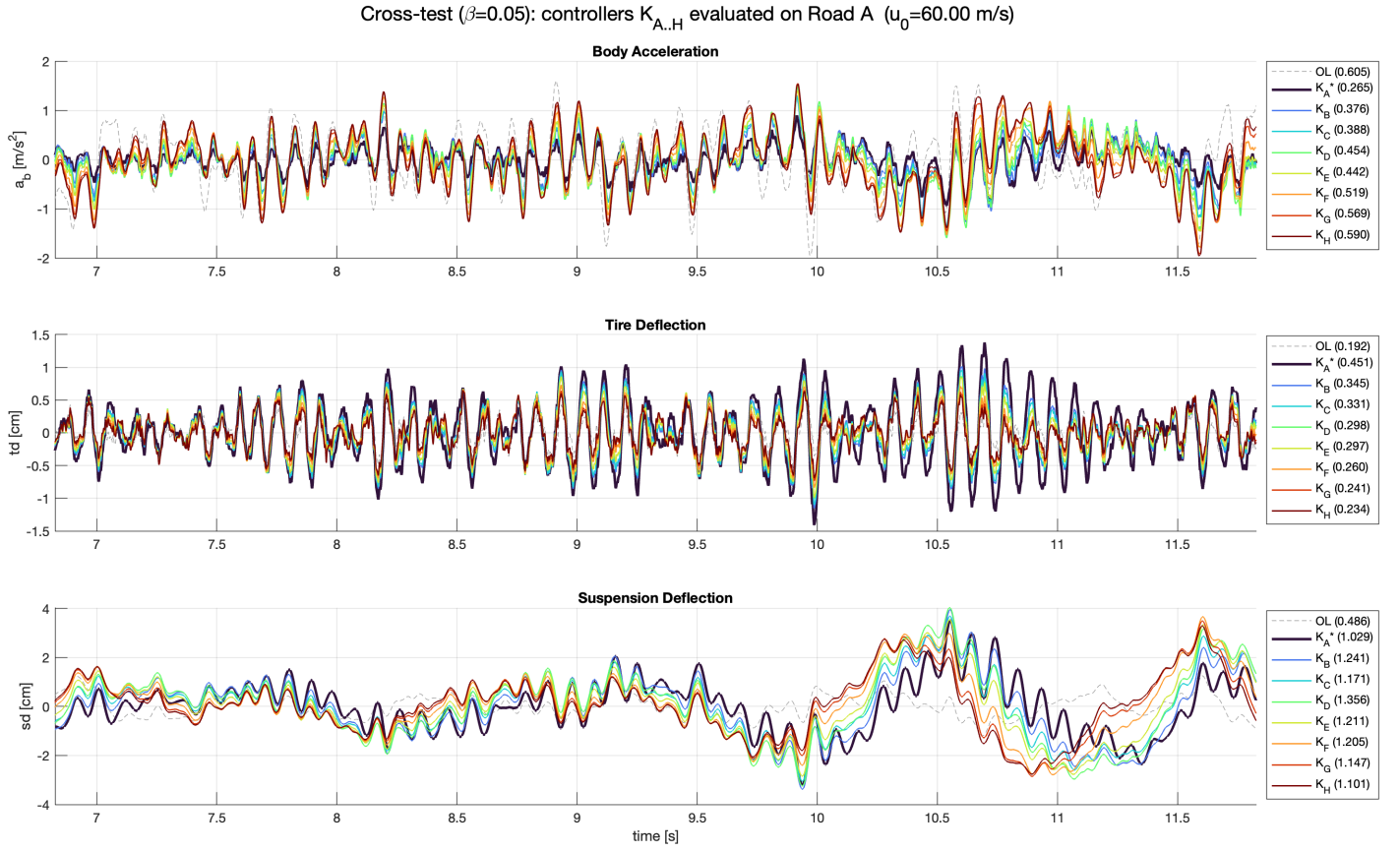


Figure 13: Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road A.

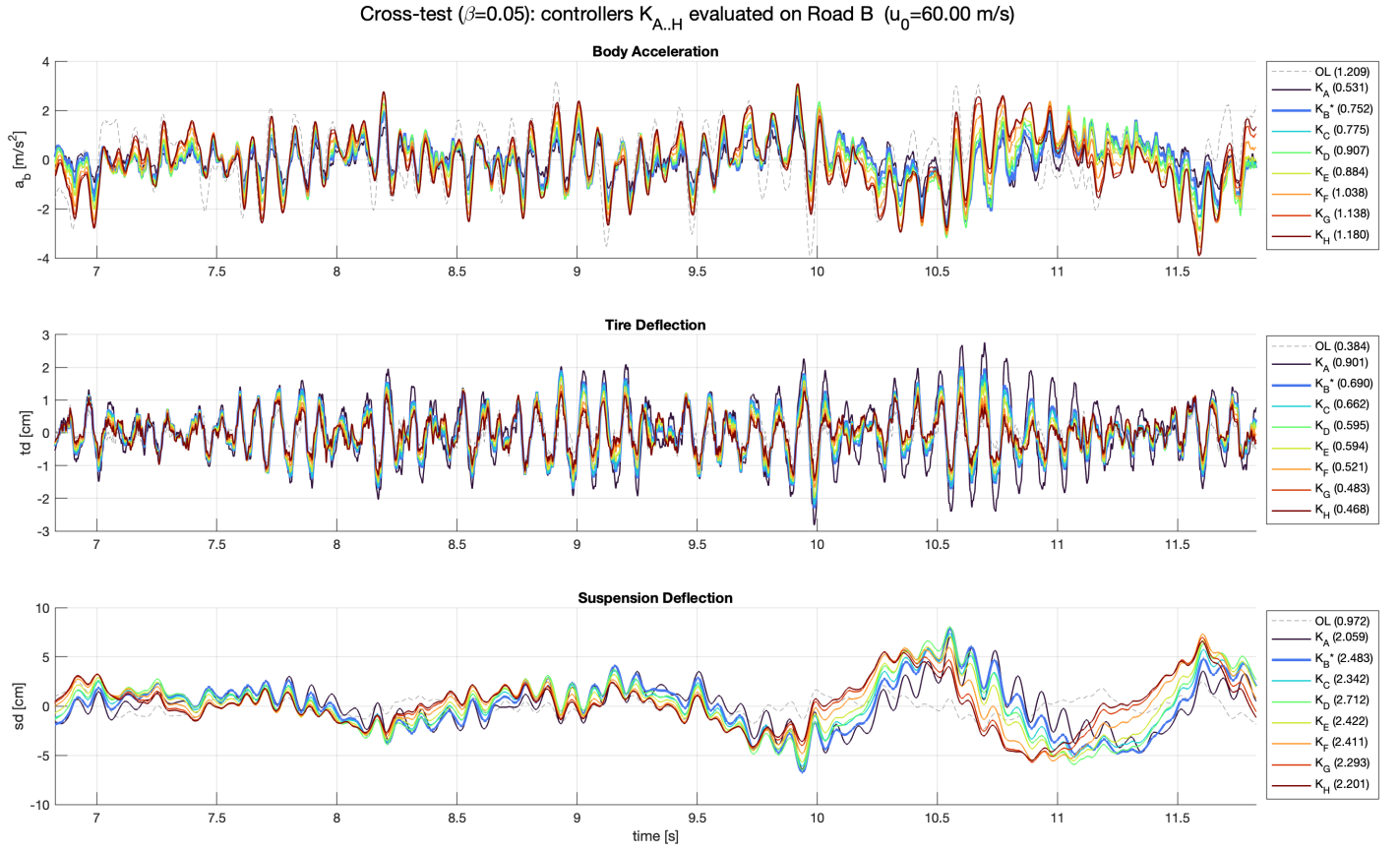


Figure 14: Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road B.

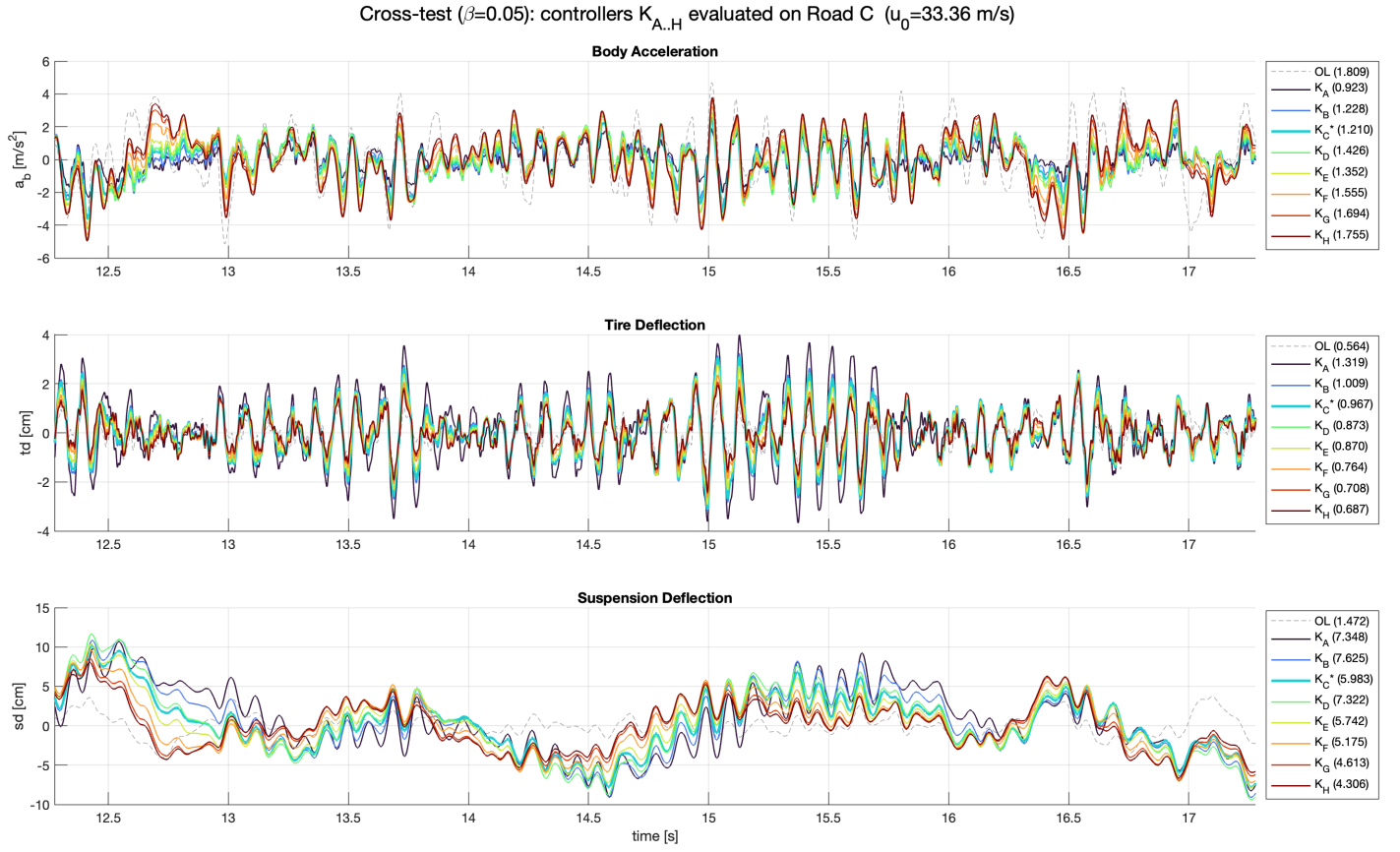


Figure 15: Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road C.

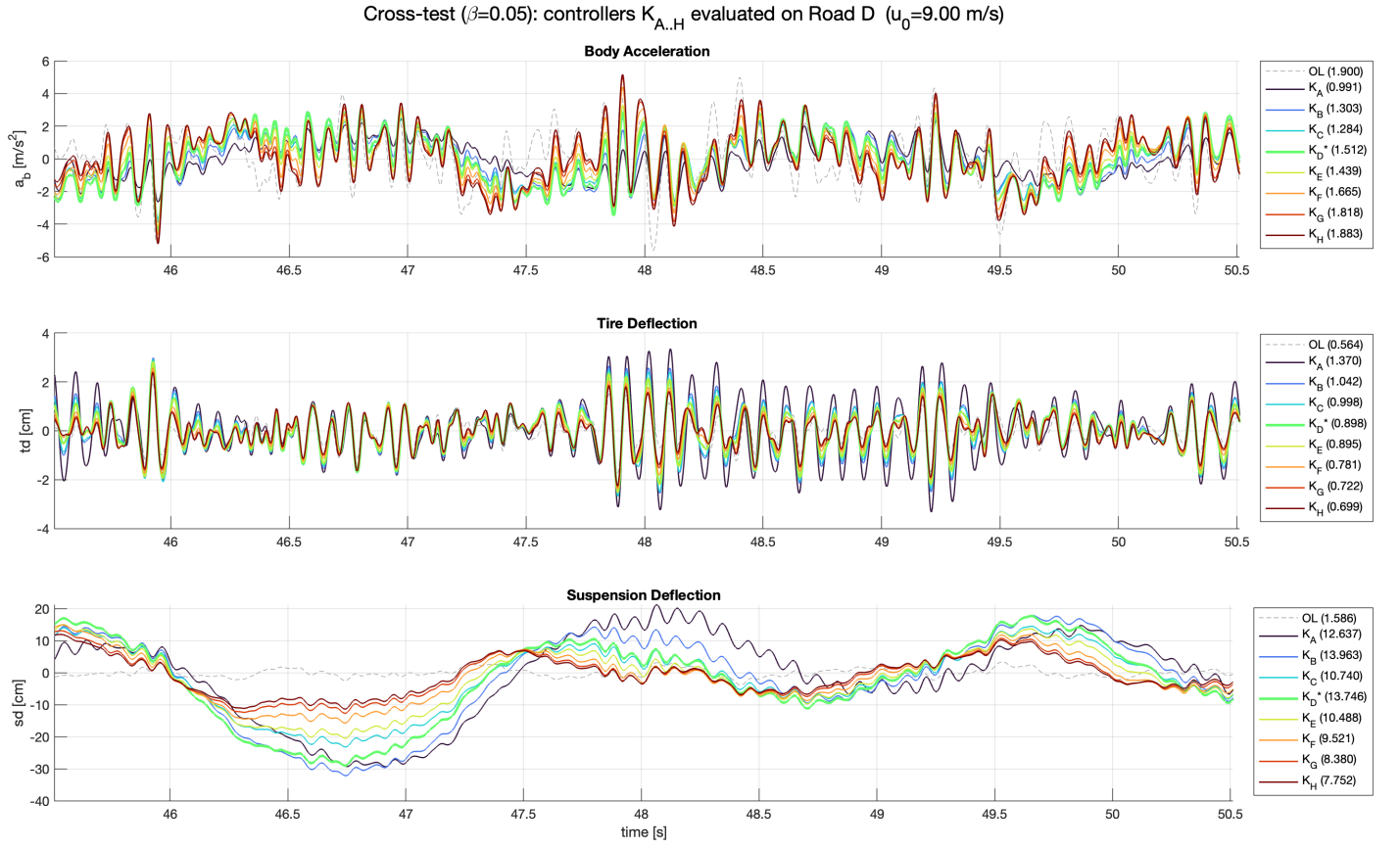


Figure 16: Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road D.

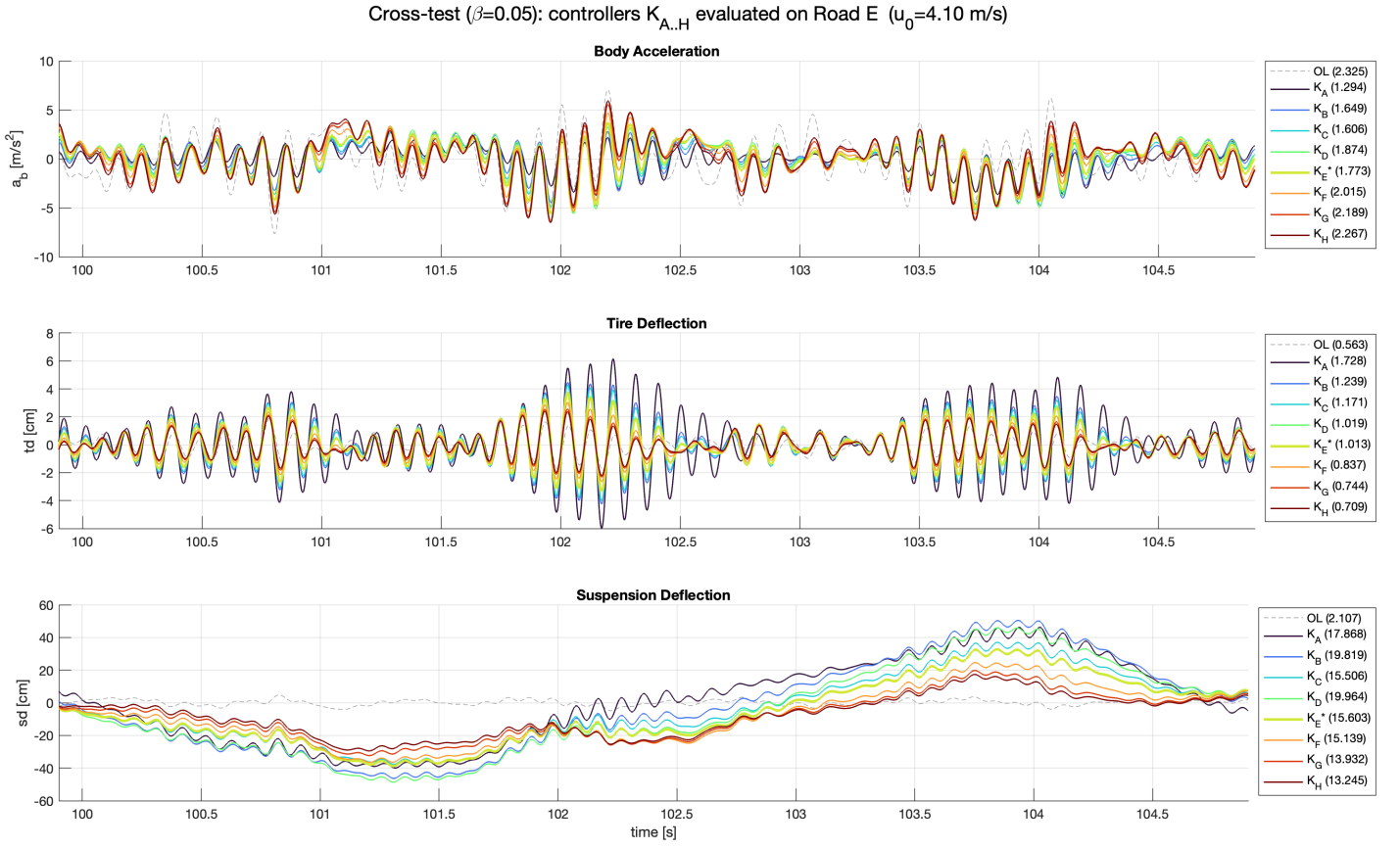


Figure 17: Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road E.

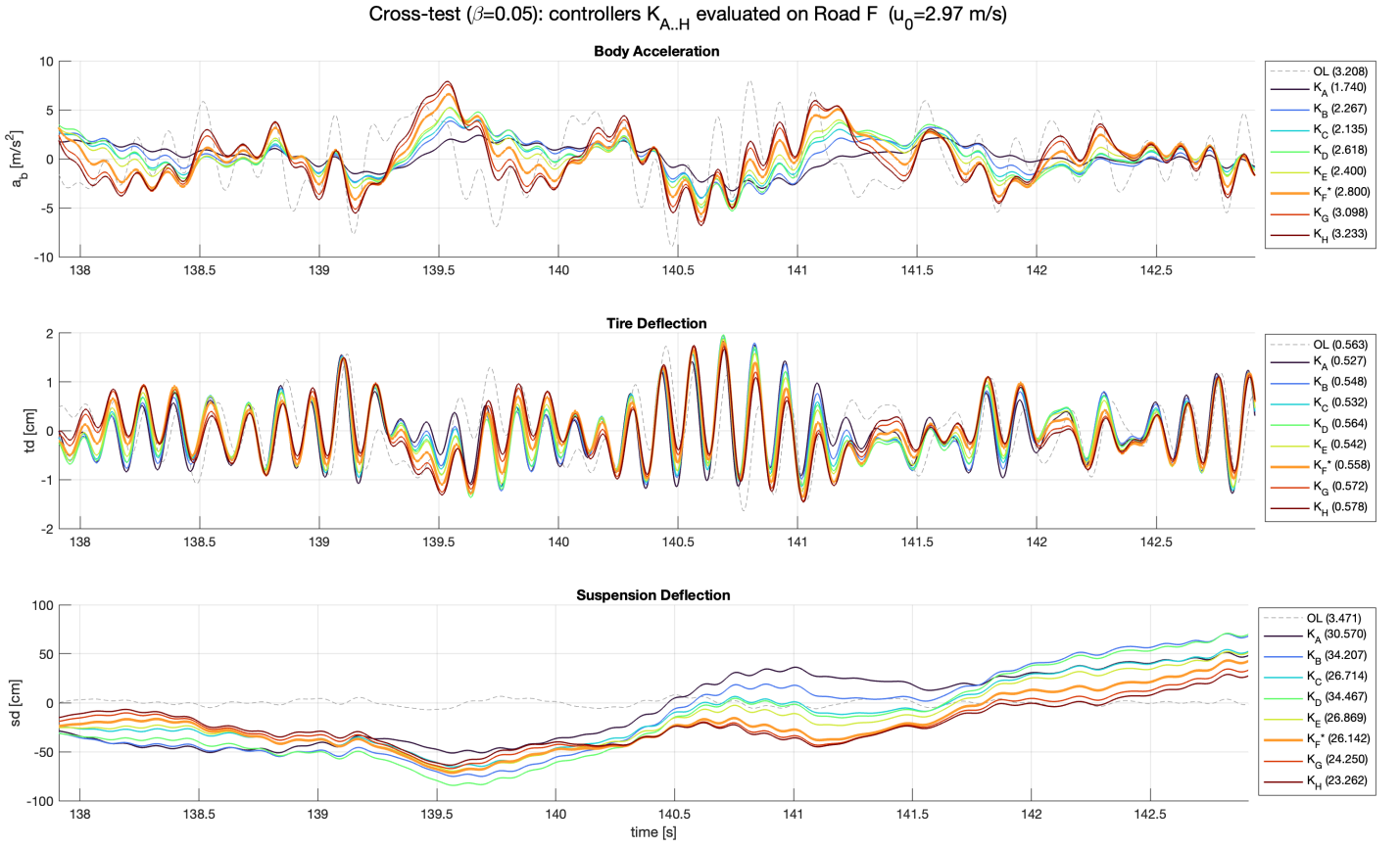


Figure 18: Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road F.

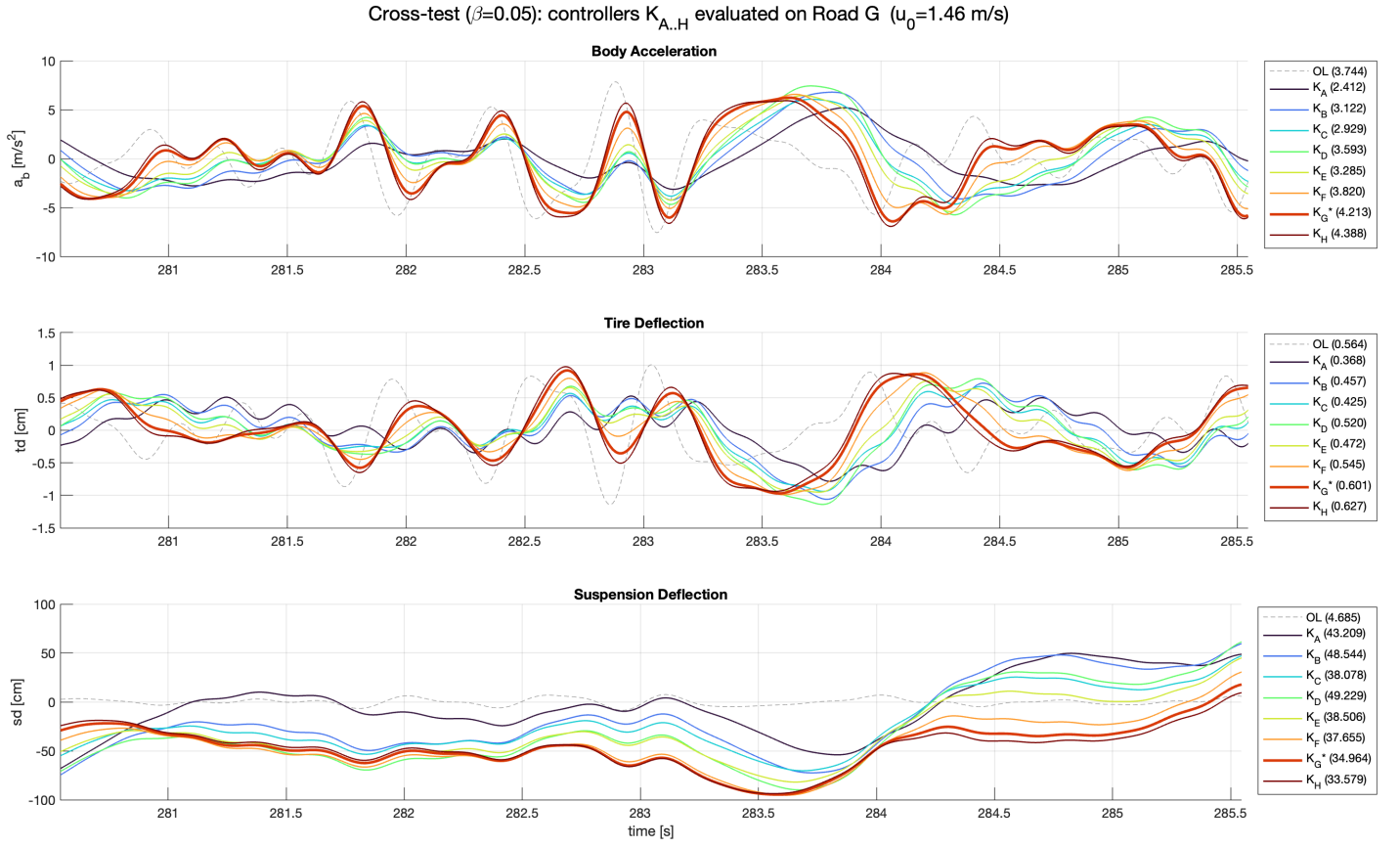


Figure 19: Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road G.

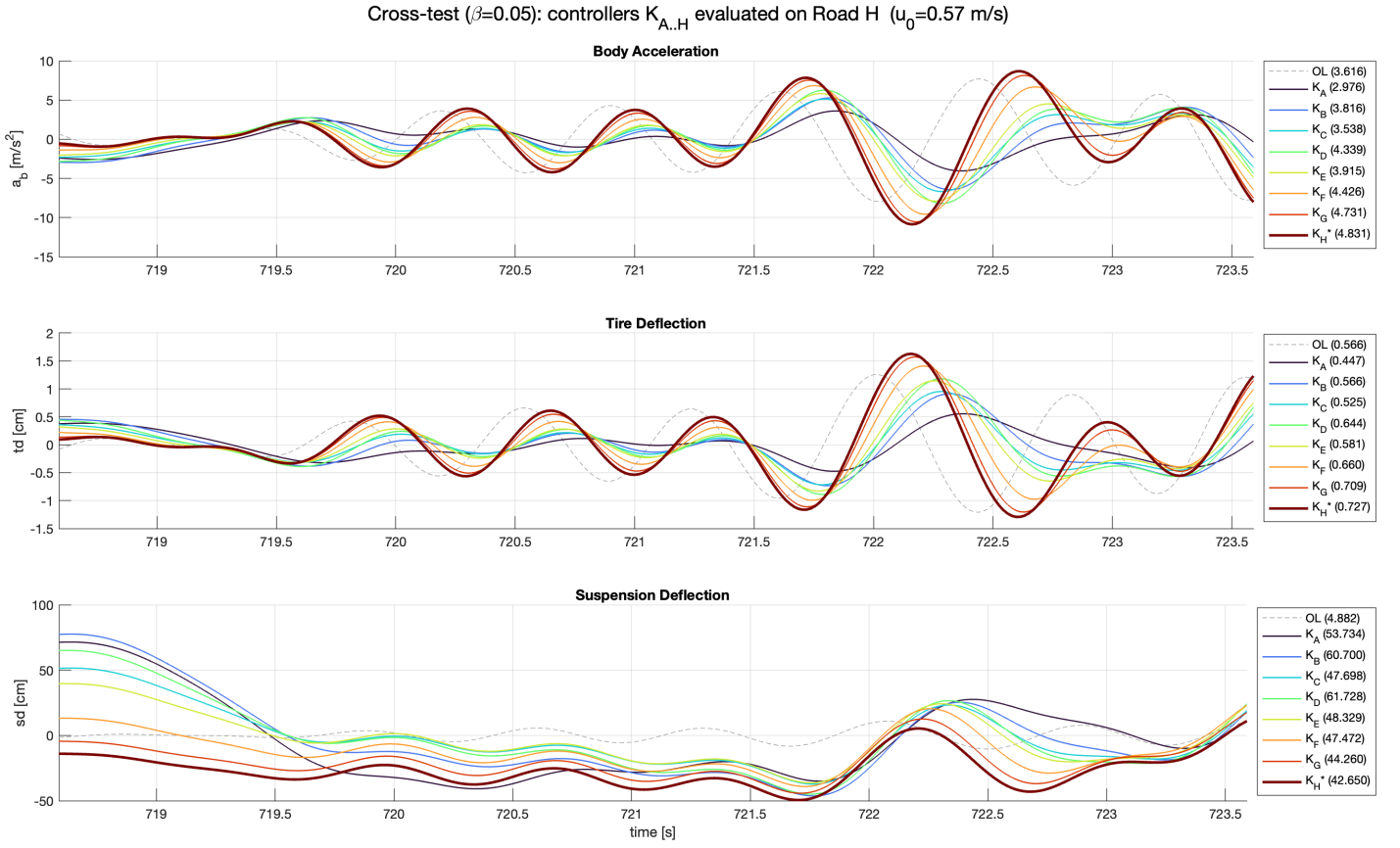


Figure 20: Cross-test performance ($\beta = 0.05$): evaluating all synthesized controllers K_{A-H} on Road H.

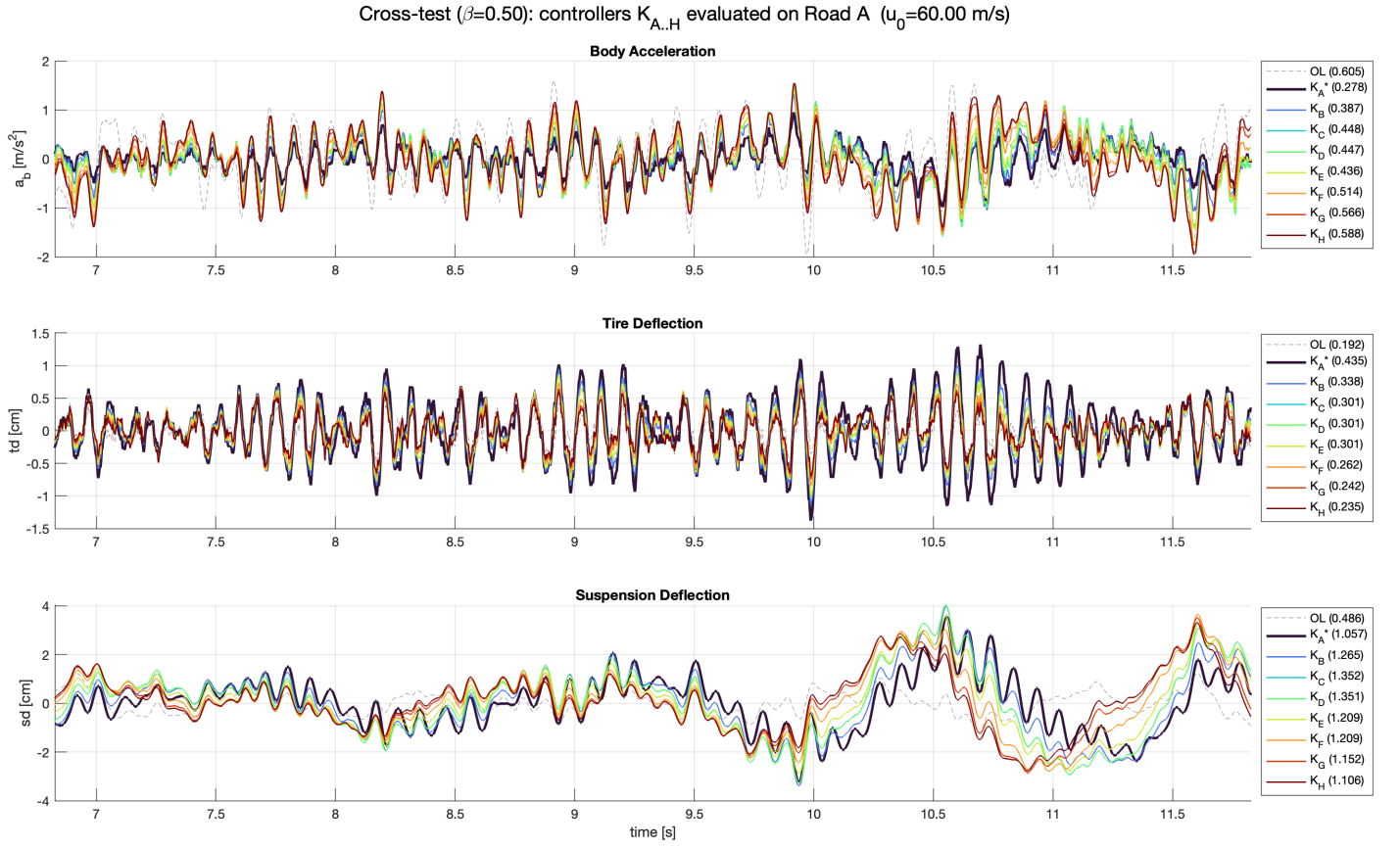


Figure 21: Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road A.

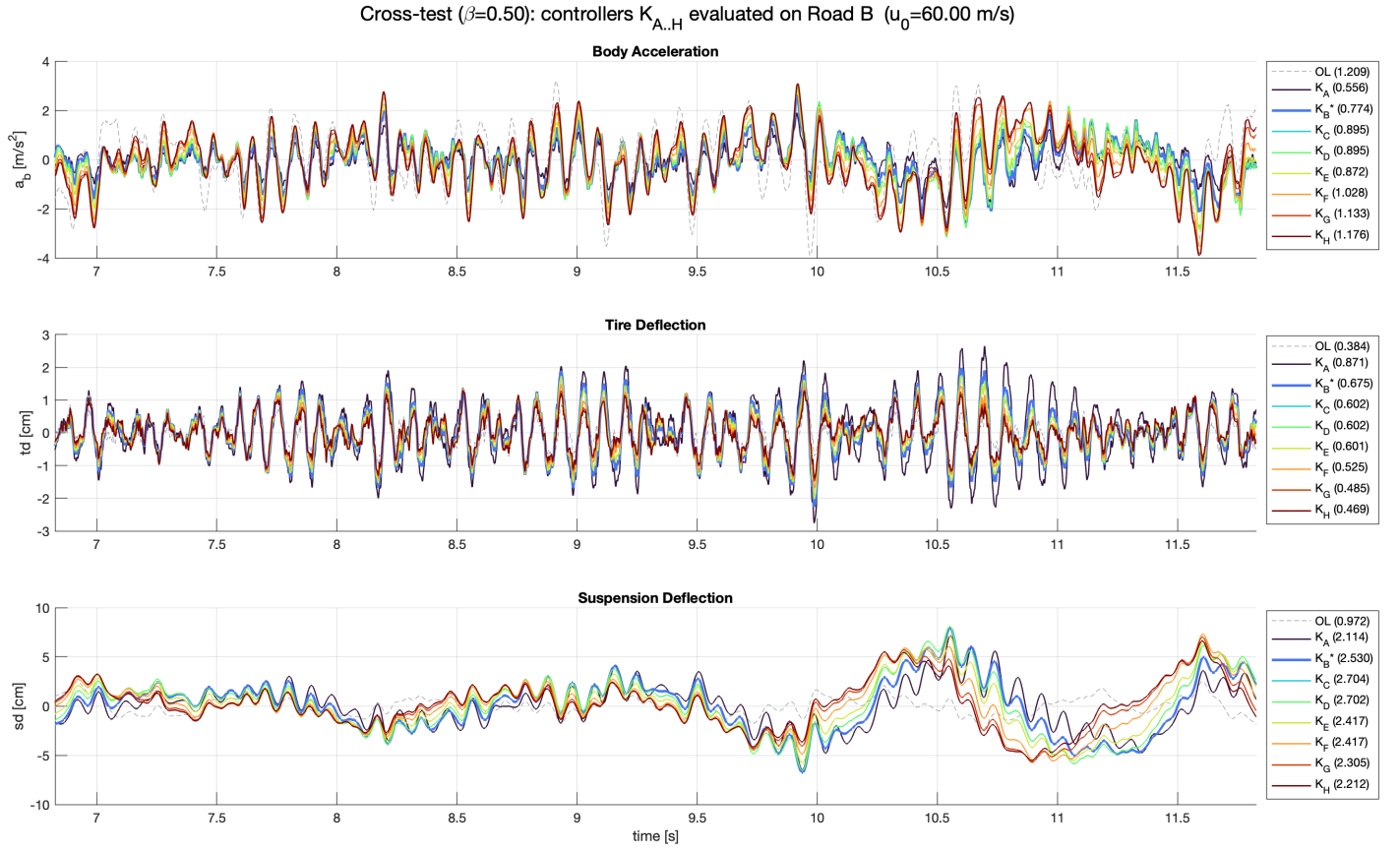


Figure 22: Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road B.

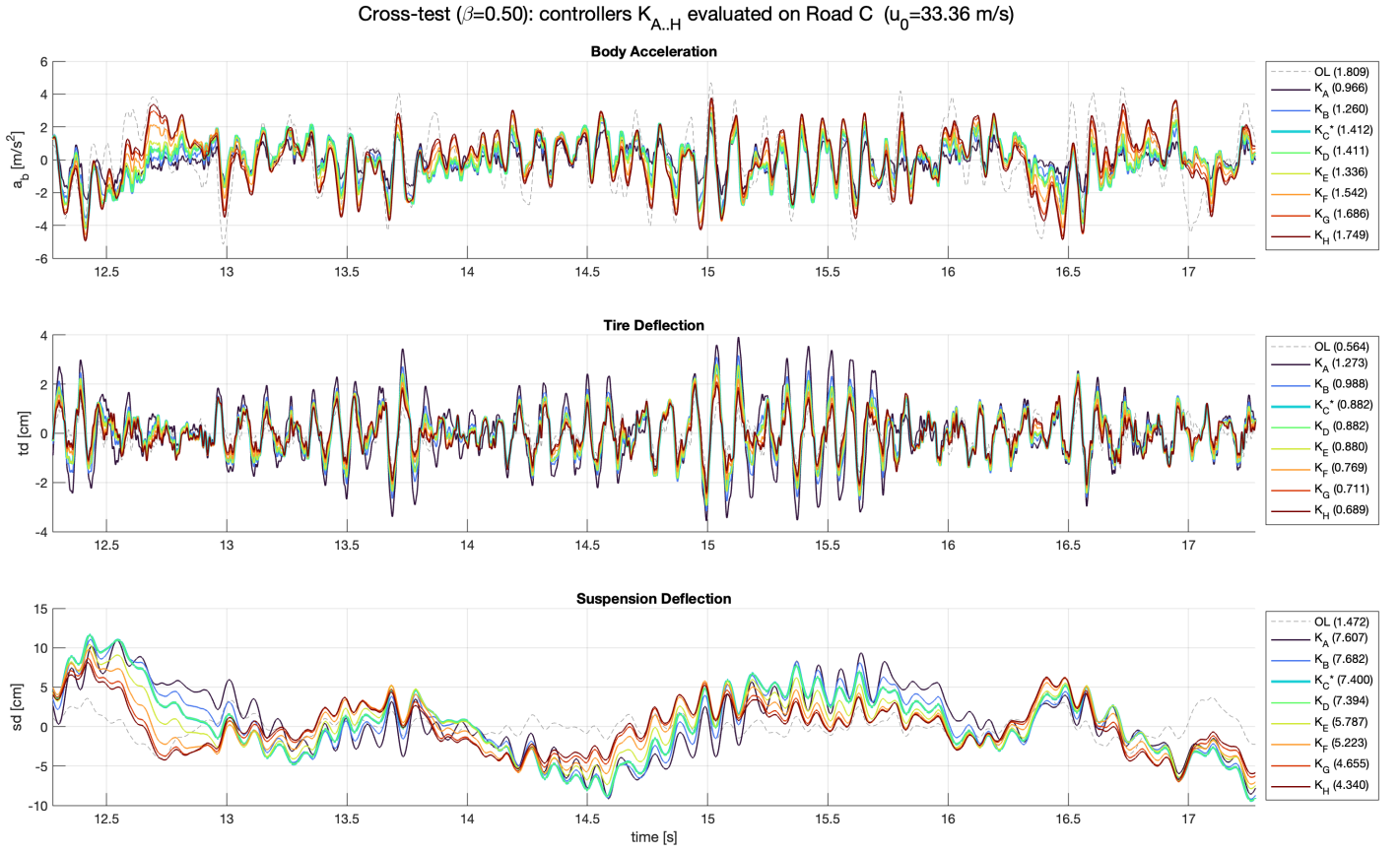


Figure 23: Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road C.

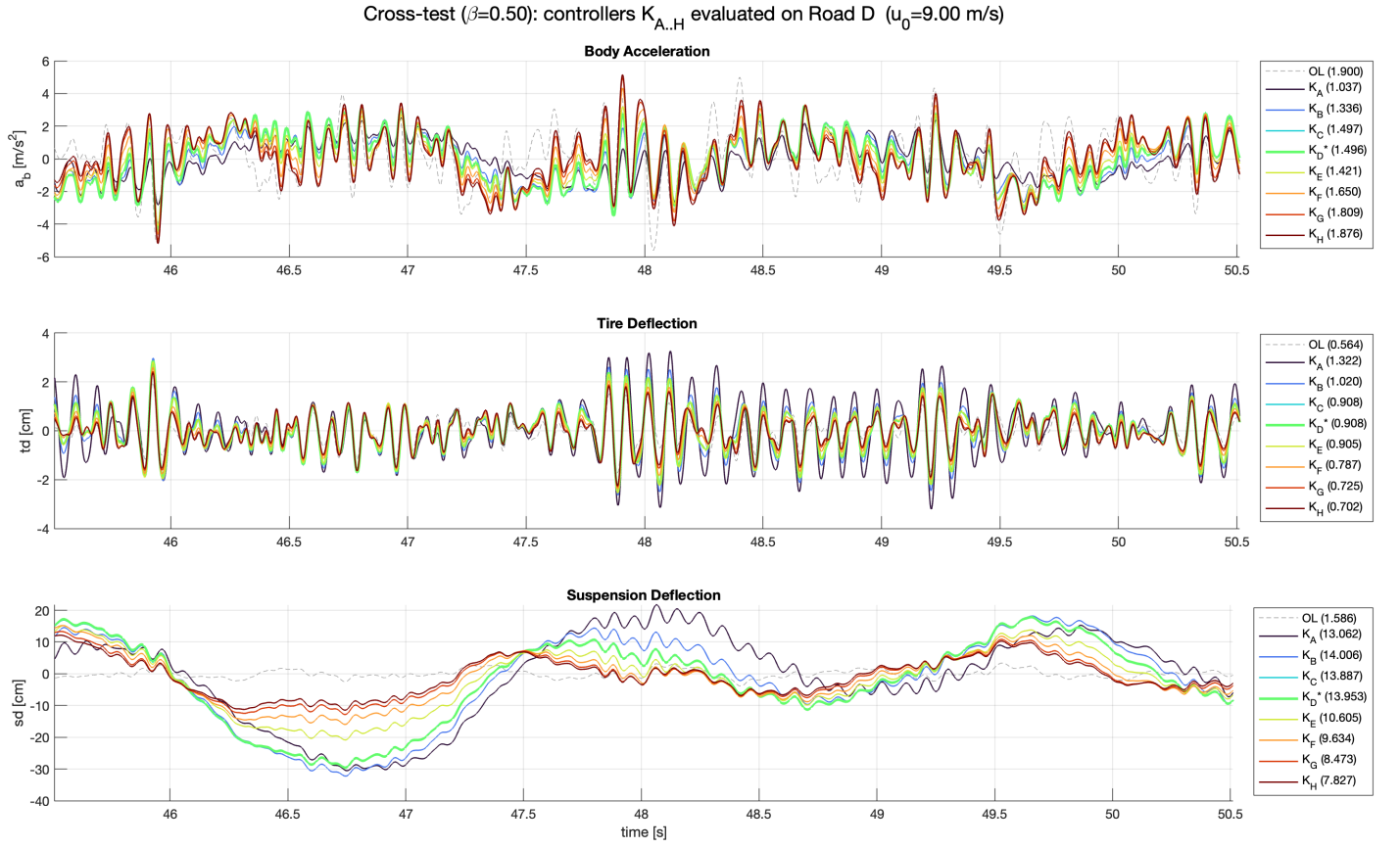


Figure 24: Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road D.

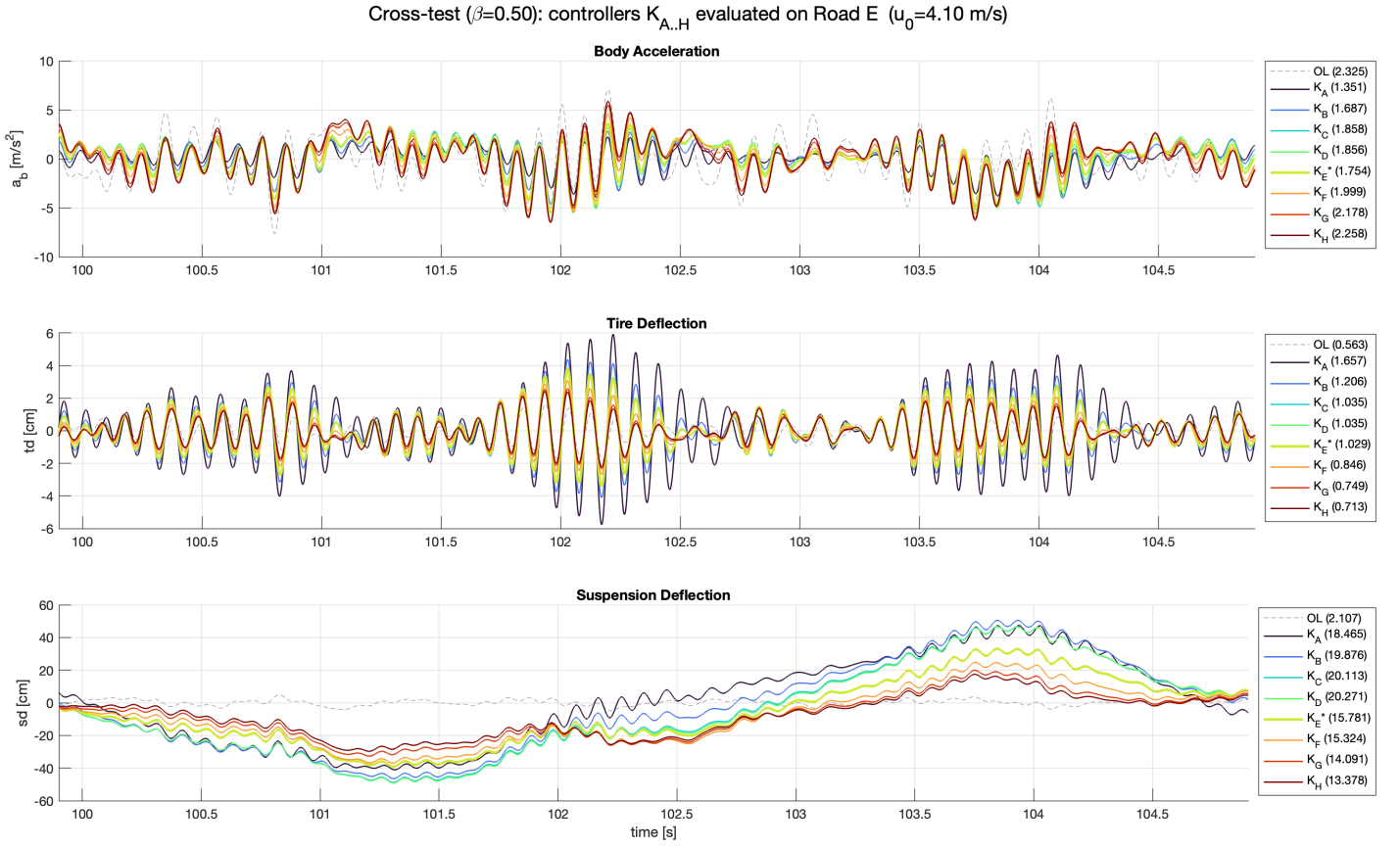


Figure 25: Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road E.

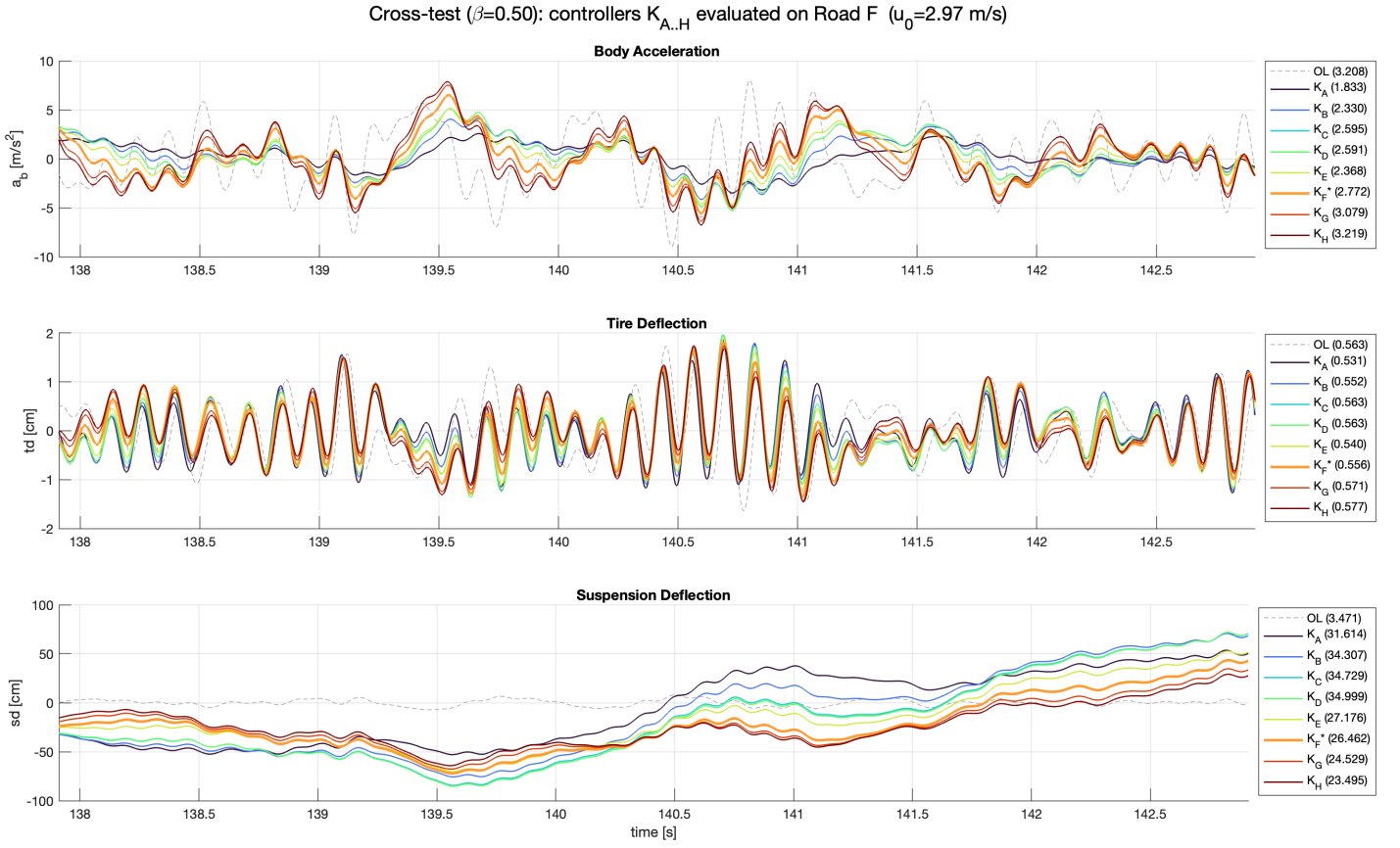


Figure 26: Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road F.

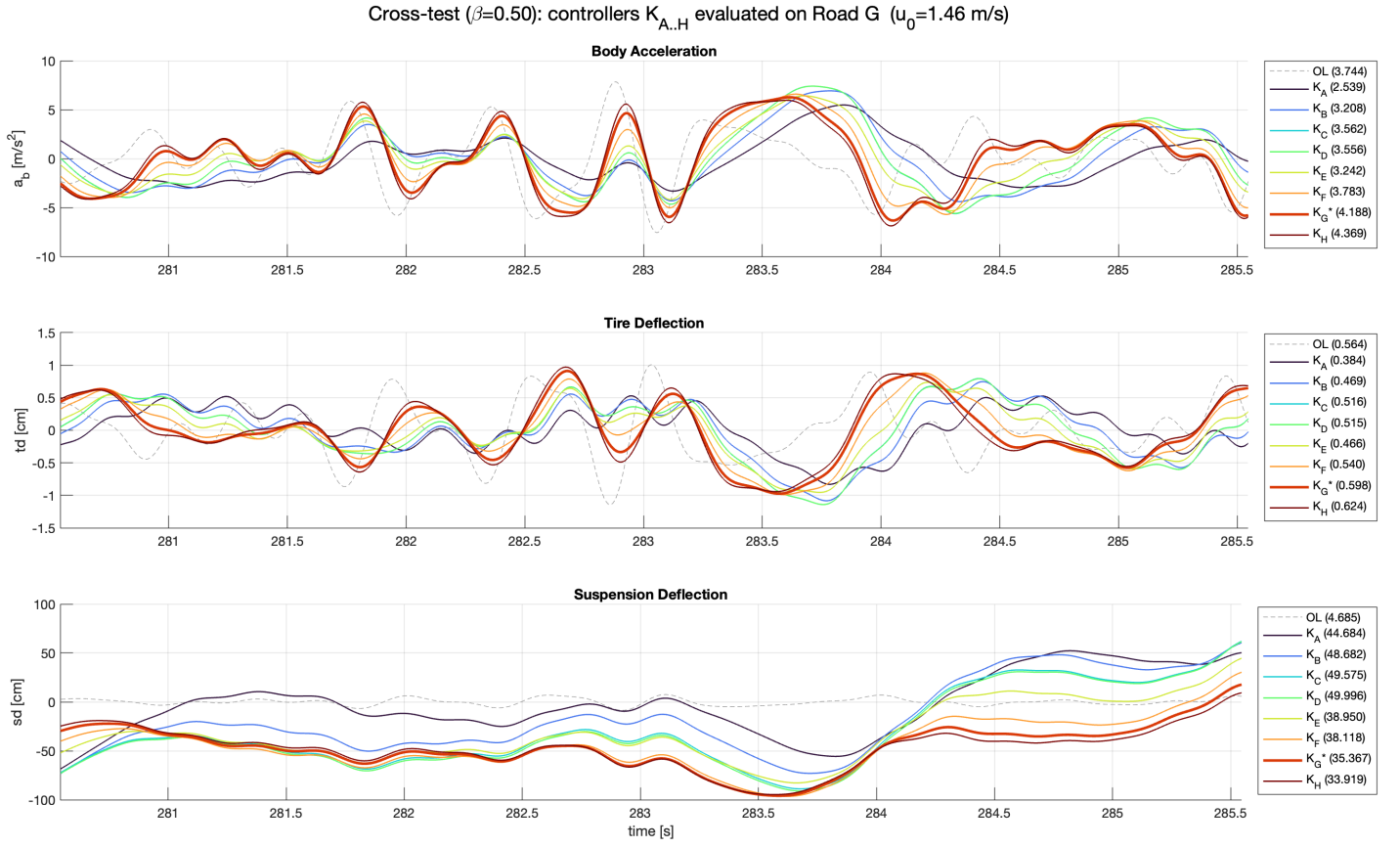


Figure 27: Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road G.

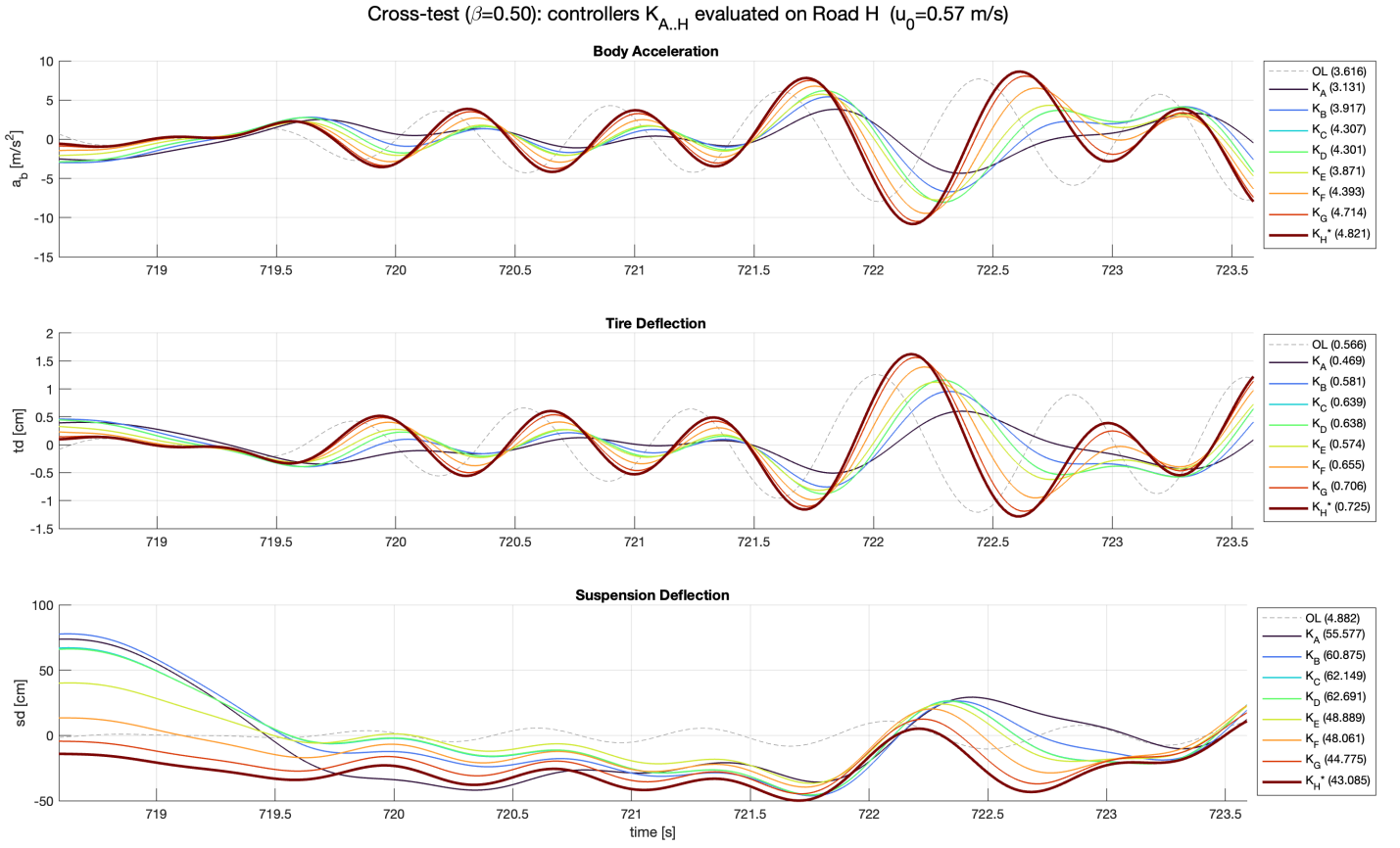


Figure 28: Cross-test performance ($\beta = 0.50$): evaluating all synthesized controllers K_{A-H} on Road H.

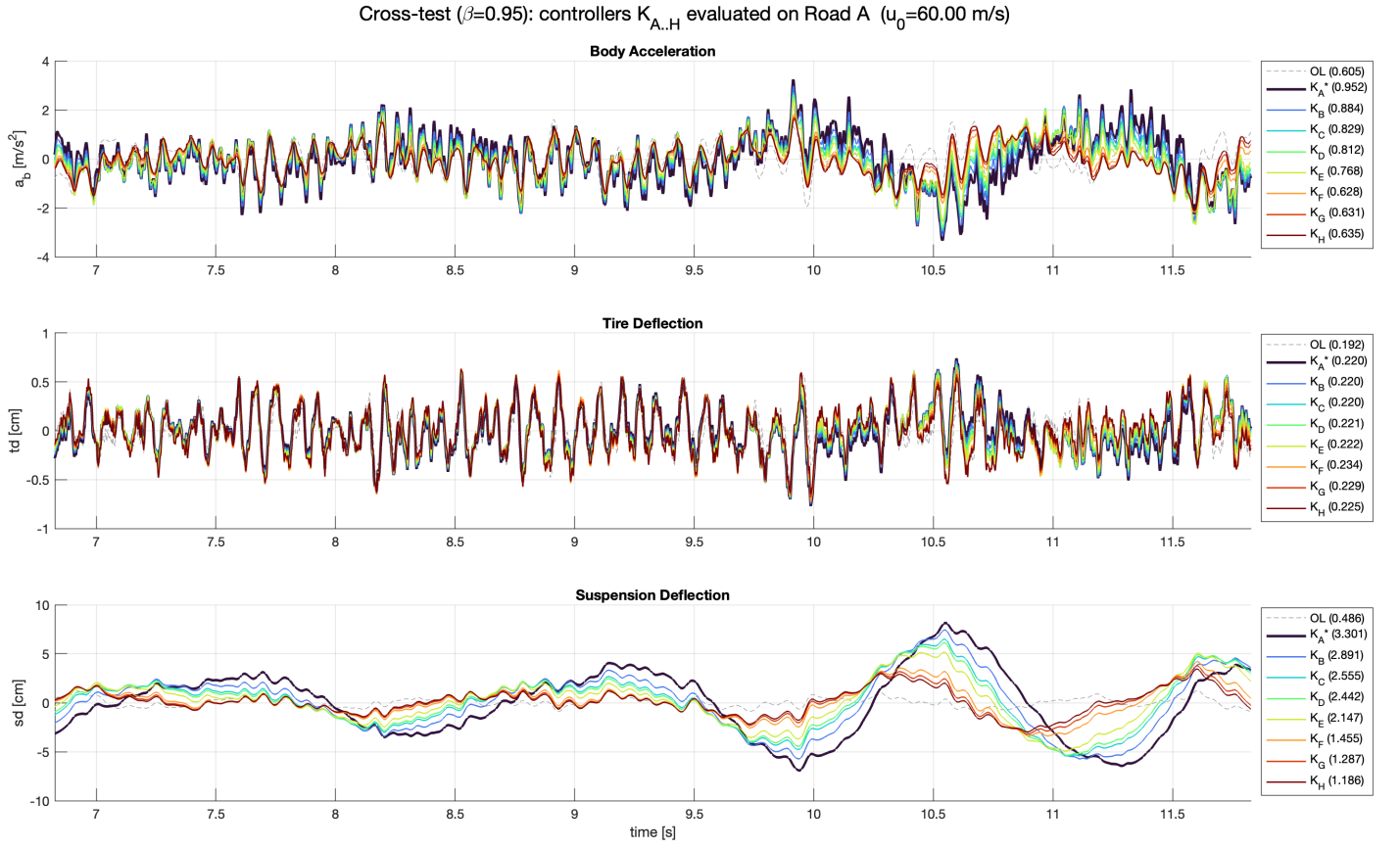


Figure 29: Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road A.

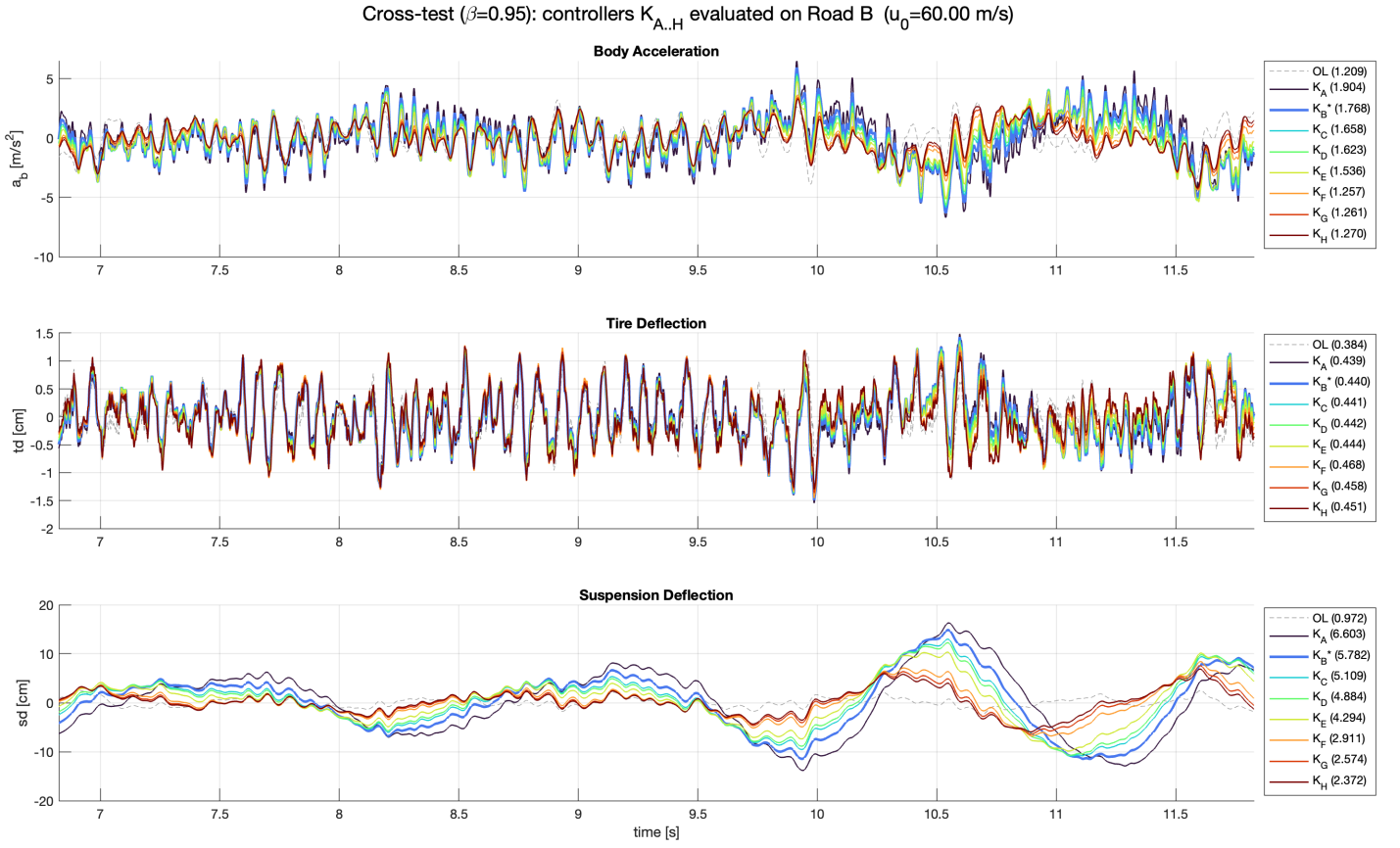


Figure 30: Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road B.

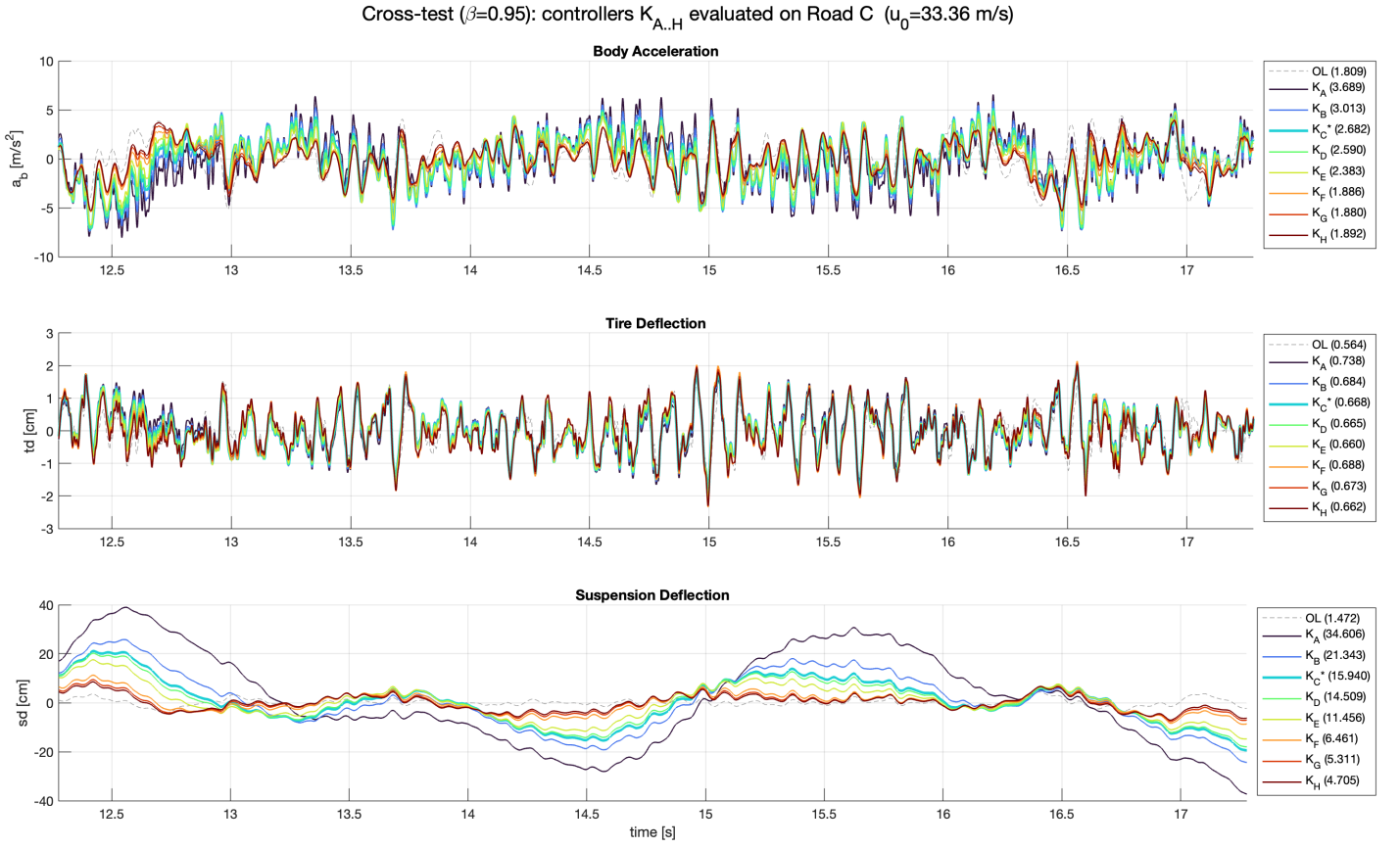


Figure 31: Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road C.

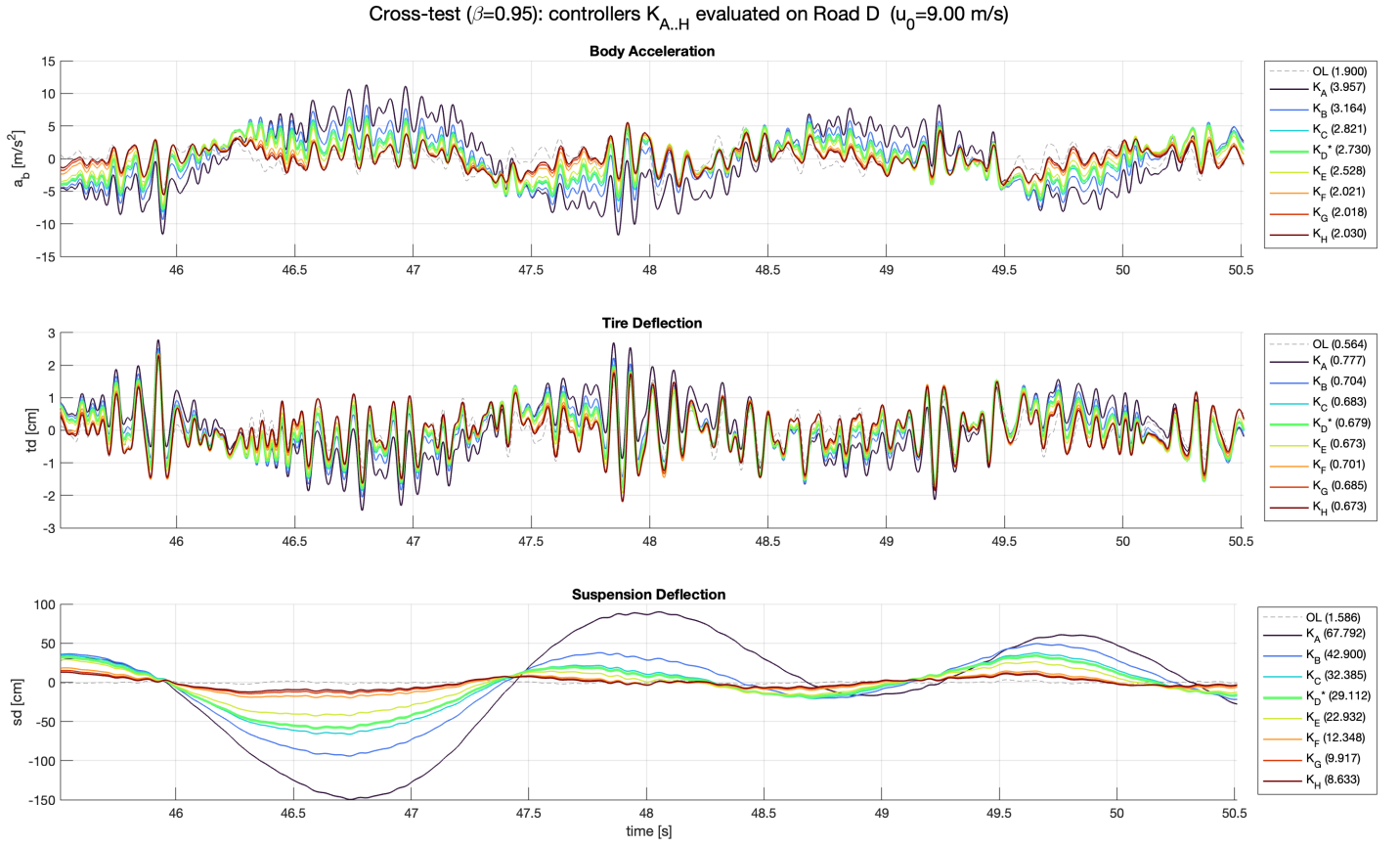


Figure 32: Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road D.

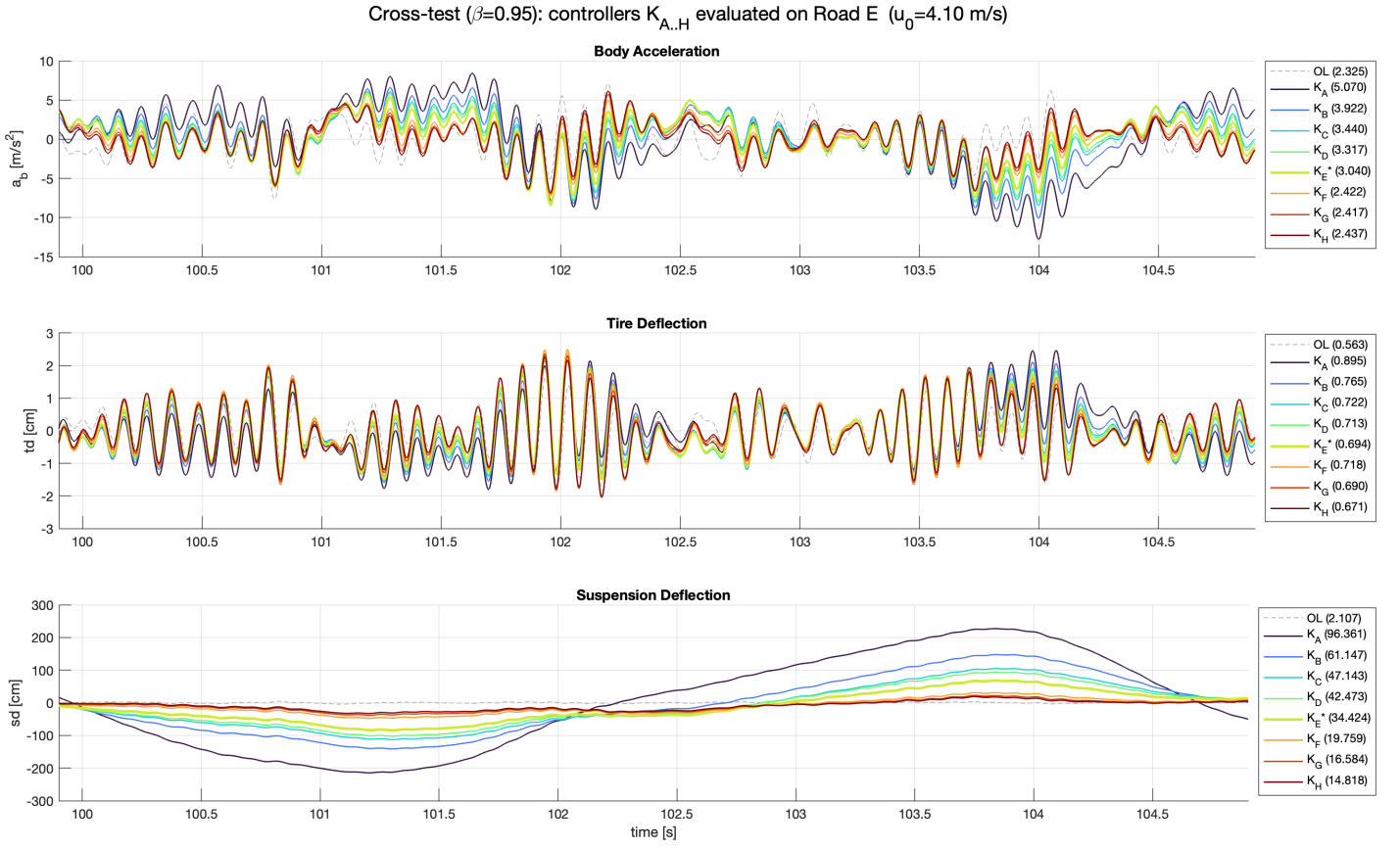


Figure 33: Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road E.

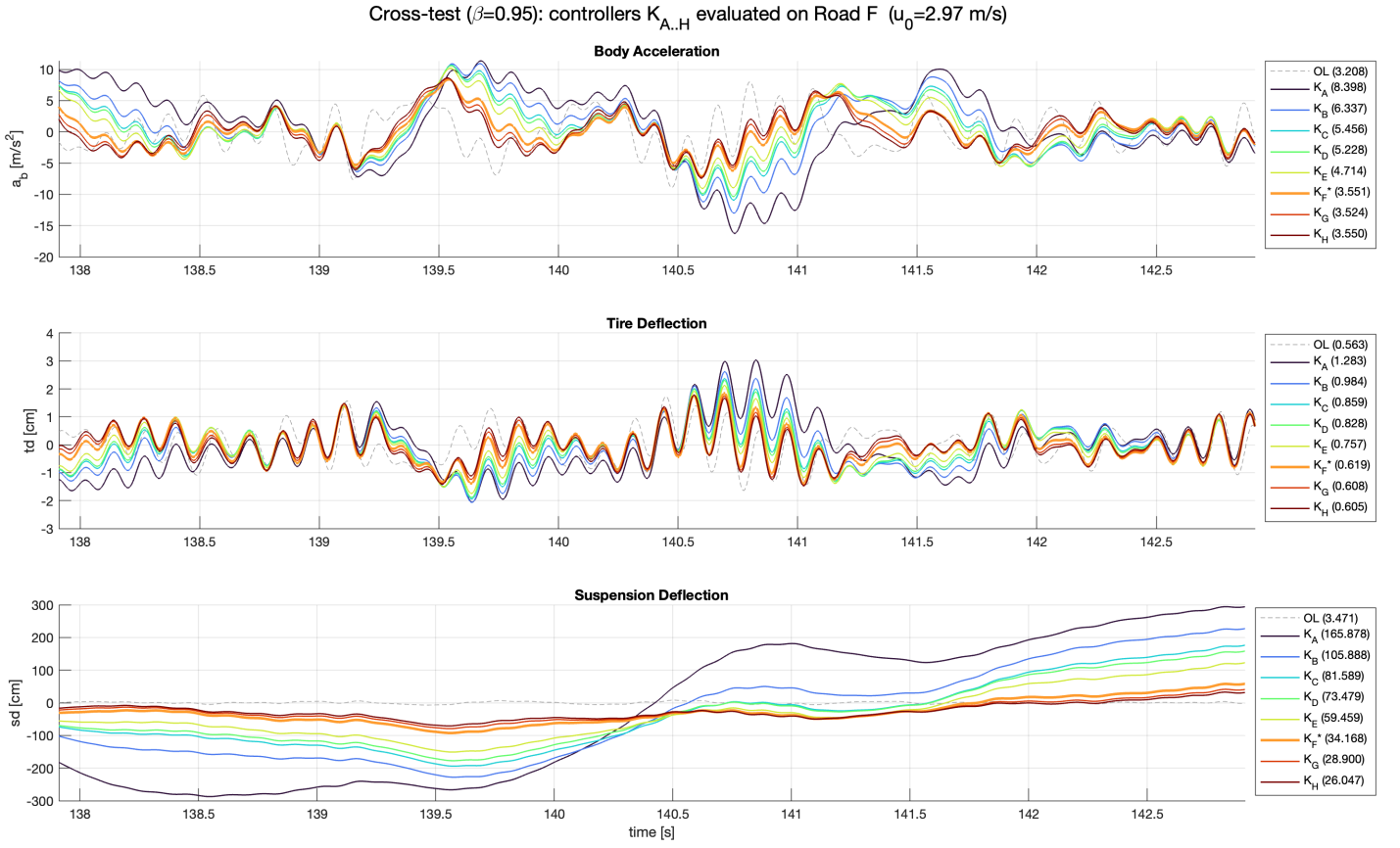


Figure 34: Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road F.

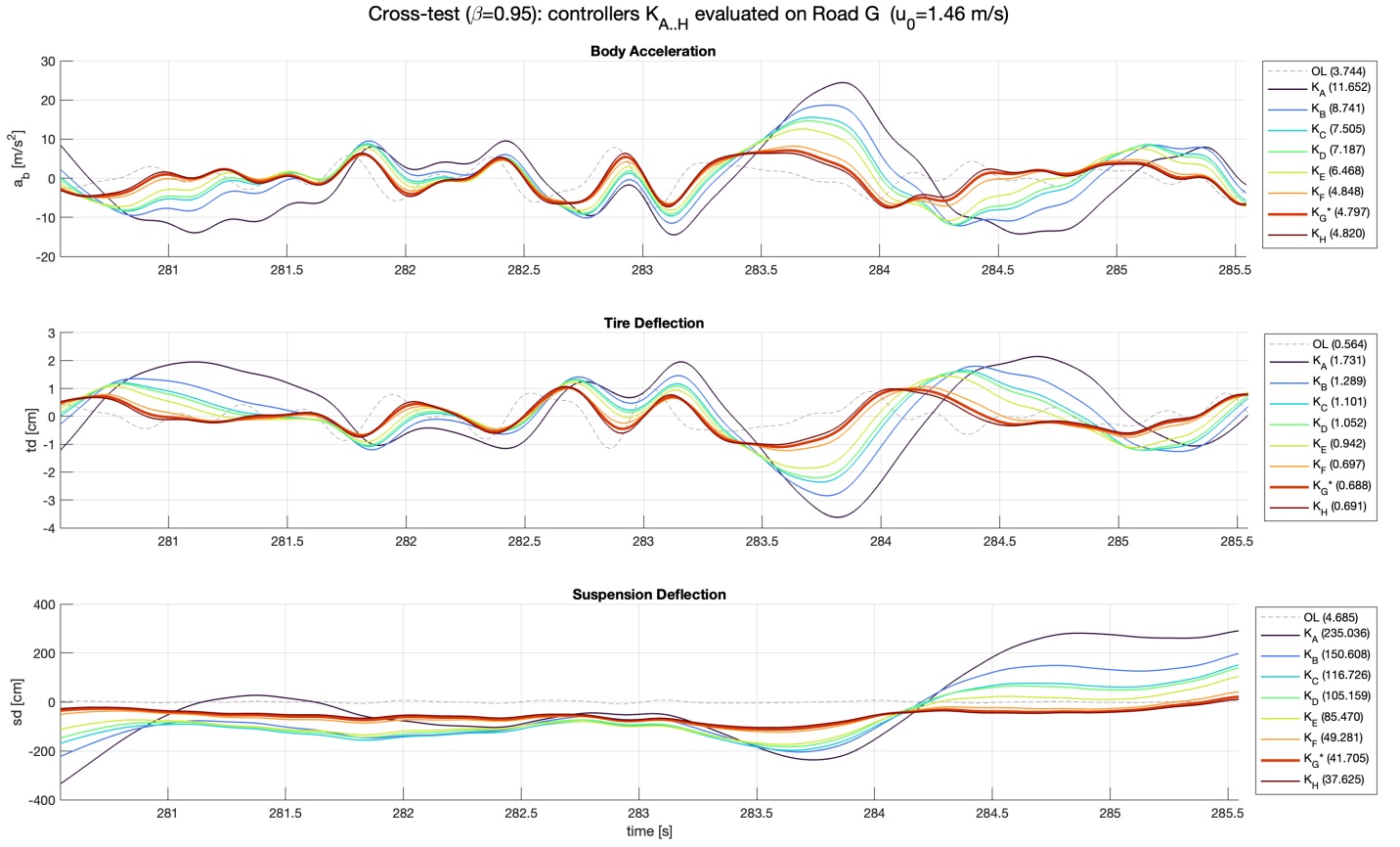


Figure 35: Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road G.

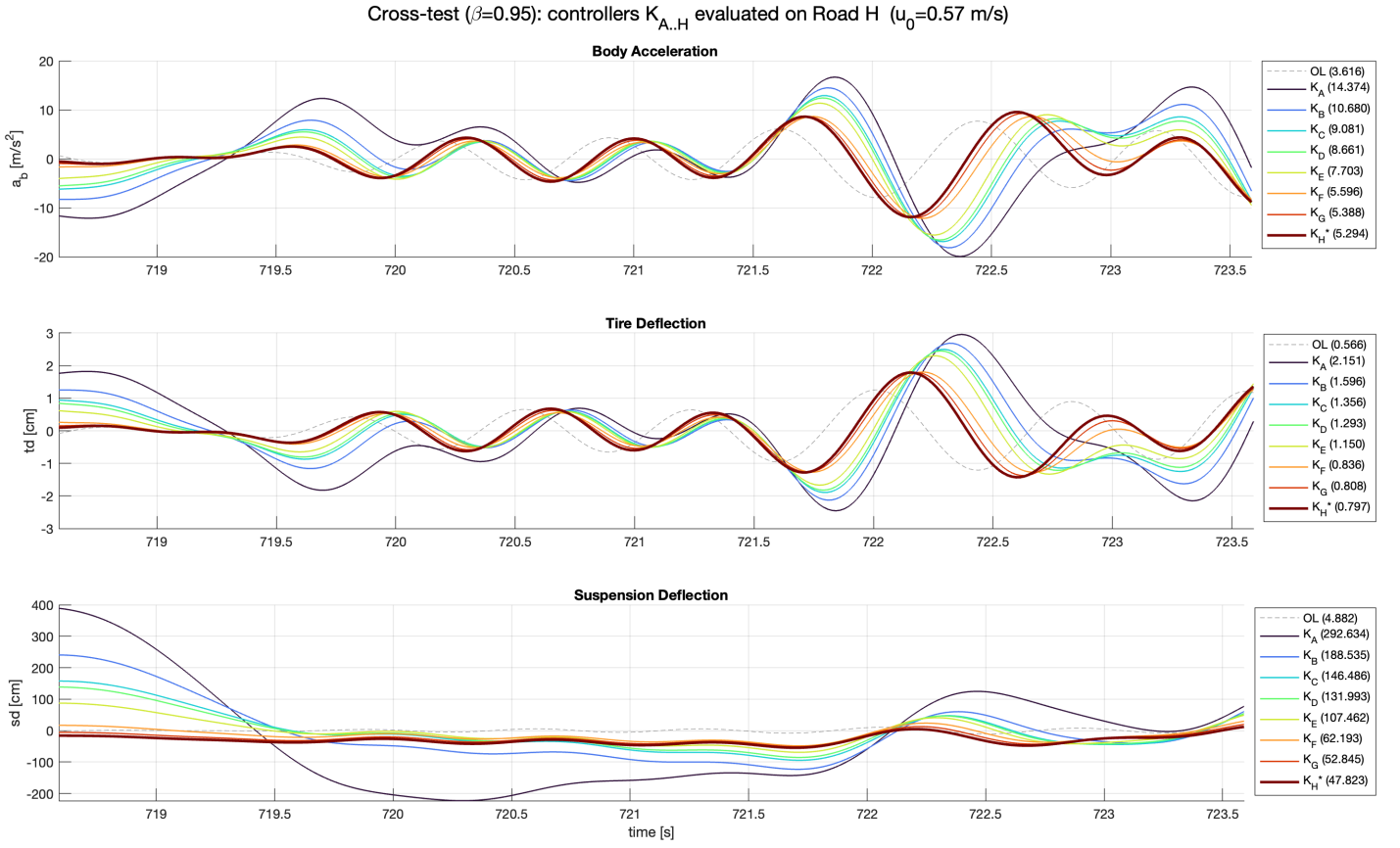


Figure 36: Cross-test performance ($\beta = 0.95$): evaluating all synthesized controllers K_{A-H} on Road H.

6.2 Codes

```
clear all; clc;
%% active suspension model used in butsuen
% for LQR control extended for h-infinity synthesis
% states:
%   x1 = zs-zu (rattle space)
%   x2 = zs_dot (body velocity)
%   x3 = zu-zr (tire deflection)
%   x4 = zu_dot (tire velocity)
% inputs:
%   u: actuator force
%   r_dot: road velocity
% outputs:
%   y1 = zs_dot_dot (body acceleration)
%   y2 = zs-zu (=x1) (rattle space)
%% init.s
% close all;
set(groot, 'DefaultFigureWindowStyle', 'docked');
%% generate road profile
road_type = 'Type D';
switch road_type
    % ISO 8608 Standard Classes
    case 'Type A', u0 = 60.00; % 3-sigma no-lift-off (capped)
    case 'Type B', u0 = 60.00; % 3-sigma no-lift-off (capped)
    case 'Type C', u0 = 33.36;
    case 'Type D', u0 = 9.00;
    case 'Type E', u0 = 4.10;
    case 'Type F', u0 = 2.97;
    case 'Type G', u0 = 1.46;
    case 'Type H', u0 = 0.57;
    otherwise
        error('Unknown road type: "%s"', road_type);
end
[road_sig, t, hsum] = roadprofile_fun(road_type, u0);
rms_road = rms(hsum);
r_iso = hsum(1:length(t)); % road displacement [m]
dt = t(2) - t(1);
rdot_raw = [diff(hsum)./dt, 0];
rdot_iso = rdot_raw;

%% define parameters
params = 'nom';
if strcmp(params,'bd')
    ms = 300; mu = 60; bs = 1000; ks = 16000; kt = 190000; bt = 0;
elseif strcmp(params,'nom')
    ms = 240; mu = 36; bs = 1000; ks = 16000; kt = 160000; bt = 0;
elseif strcmp(params,'suv')
```

```

    ms = 600; mu = 60; bs = 3000; ks = 20000; kt = 180000; bt = 0;
elseif strcmp(params,'truck')
    ms = 4000; mu = 200; bs = 8000; ks = 150000; kt = 800000; bt = 0;
elseif strcmp(params,'sunlusoy')
    ms = 250; mu = 35; bs = 800; ks = 15000; kt = 120000; bt = 0;
end

% system dynamics
A = [0, 1, 0, -1;...
    -ks/ms, -bs/ms, 0, bs/ms;...
    0, 0, 0, 1;...
    ks/mu, bs/mu, -kt/mu, -(bs+bt)/mu];
B = [0; 1/ms; 0; -1/mu];
L = [0; 0; -1; 0];
B_x = [B,L];
C = [A(2,:); 1, 0, 0, 0; 0, 0, 1, 0];
D = [1/ms, 0; 0, 0; 0, 0];

quartercar = ss(A, B_x, C, D);
quartercar.StateName = {'rattle space'; 'body velocity'; 'tire deflection'; 'tire vel
quartercar.InputName = {'fs','r_dot'};
quartercar.OutputName = {'body acceleration'; 'rattle space'; 'tire deflection'};
quartercar_openloop = quartercar(:, 'r_dot');

%% time simulation for open-loop
[y_ol_a,~] = lsim(quartercar('body acceleration','r_dot'), rdot_iso, t);
[y_ol_s,~] = lsim(quartercar('rattle space','r_dot'), rdot_iso, t);
rms_a = rms(y_ol_a);
rms_s = rms(y_ol_s);
rms_r = rms(rdot_iso);

%% lqr design
ro1 = 0.4; ro2 = 0.16; ro3 = 0.4; ro4 = 0.16;
Q = [ks^2/ms^2+ro1, bs*ks/ms^2, 0, -bs*ks/ms^2;...
    bs*ks/ms^2, bs^2/ms^2+ro2, 0, -bs^2/ms^2;...
    0, 0, ro3, 0;...
    -bs*ks/ms^2, -bs^2/ms^2, 0, bs^2/ms^2+ro4];
N = [-ks/ms^2; -bs/ms^2; 0; bs/ms^2];
R = 1/ms^2;
K_lqr = lqr(A, B, Q, R, N);
A_cl = A-B*K_lqr;
C_cl = [A_cl(2,:); 1, 0, 0, 0; 0, 0, 1, 0];
quartercar_cl_lqr = ss(A-B*K_lqr,L,C_cl,D(:,2));
quartercar_cl_lqr.StateName = {'rattle space'; 'body velocity'; 'tire deflection'; 't
quartercar_cl_lqr.InputName = {'r_dot'};
quartercar_cl_lqr.OutputName = {'body acceleration'; 'rattle space'; 'tire deflection

% temp lqr analysis

```

```

A_cl = A - B*K_lqr;
C_u = -K_lqr;
D_u = 0;
sys_lqr_u = ss(A_cl, L, C_u, D_u);
sys_lqr_u.InputName = 'r_dot';
sys_lqr_u.OutputName = 'u';
[u_lqr, ~] = lsim(sys_lqr_u, rdot_iso, t);

fprintf('--- LQR control force ---\n');
fprintf(' RMS:  %.2f N\n', rms(u_lqr));
fprintf(' Peak:  %.2f N\n', max(abs(u_lqr)));

%% hinf design
[quartercar_cl_hinf, K, gamma, qcar,...
 info, rdot_iso, r_iso] = ...
 design_hinf_for_road('Type A',...
 1,1,3, 0.05); % synthesize A

%% open-loop vs lqr bode
figure(1); clf;
opts = bodeoptions;
opts.FreqUnits = 'Hz'; opts.MagScale = 'log';
opts.MagUnits = 'abs'; opts.YLim = [10^-6, 10^2];
bodemag(quartercar_openloop, 'b--', opts); hold on
bodemag(quartercar_cl_lqr, 'r', opts);
grid minor
legend('Open-loop', 'LQR', 'location', 'SouthEast')
title('Open-loop vs LQR Bode');
xlim([0.01, 1000])

%% open-loop vs h-infinity bode
figure(2); clf;
opts = bodeoptions;
opts.FreqUnits = 'Hz'; opts.MagScale = 'log';
opts.MagUnits = 'abs'; opts.YLim = [10^-6, 10^3];
bodemag(qcar(['ab'; 'sd'; 'td'], 'r'), 'b--', opts); hold on
bodemag(quartercar_cl_hinf(['ab'; 'sd'; 'td'], 'r'), 'k', opts);
grid minor
legend('Open-loop', 'H-infinity', 'location', 'SouthEast')
title('Open-loop vs H-infiinty Bode');
xlim([0.01, 1000])

%% time simulations
[y_ab_ol, ~] = lsim(quartercar('body acceleration', 'r_dot'), rdot_iso, t);
[y_ab_lqr, ~] = lsim(quartercar_cl_lqr('body acceleration', 'r_dot'), rdot_iso, t);
[y_ab_hinf, ~] = lsim(quartercar_cl_hinf('ab', 'r'), r_iso, t);
[y_td_ol, ~] = lsim(quartercar('tire deflection', 'r_dot'), rdot_iso, t);
[y_td_lqr, ~] = lsim(quartercar_cl_lqr('tire deflection', 'r_dot'), rdot_iso, t);

```

```

[y_td_hinf, ~] = lsim(quartercar_cl_hinf('td','r'), r_iso, t);
[y_rs_ol, ~] = lsim(quartercar('rattle space','r_dot'), rdot_iso, t);
[y_rs_lqr, ~] = lsim(quartercar_cl_lqr('rattle space','r_dot'), rdot_iso, t);
[y_rs_hinf, ~] = lsim(quartercar_cl_hinf('sd','r'), r_iso, t);

figure(3); clf;
subplot(3,1,1);
plot(t, y_ab_ol, 'b--','LineWidth',0.8); hold on;
plot(t, y_ab_lqr, 'r', 'LineWidth',1.0);
plot(t, y_ab_hinf,'k', 'LineWidth',1.0);
title('Body Acceleration (Comfort)');
ylabel('m/s^2'); grid on;
legend(sprintf('OL (%.4f)', rms(y_ab_ol)), ...
        sprintf('LQR (%.4f)', rms(y_ab_lqr)), ...
        sprintf('H-inf (%.4f)',rms(y_ab_hinf)));
xlim([10, 15]);
subplot(3,1,2);
plot(t, y_td_ol*100, 'b--','LineWidth',0.8); hold on;
plot(t, y_td_lqr*100, 'r', 'LineWidth',1.0);
plot(t, y_td_hinf*100,'k', 'LineWidth',1.0);
title('Tire Deflection');
ylabel('cm'); xlabel('Time (s)'); grid on;
legend(sprintf('OL (%.4f)', rms(y_rs_ol)*100), ...
        sprintf('LQR (%.4f)', rms(y_rs_lqr)*100), ...
        sprintf('H-inf (%.4f)',rms(y_rs_hinf)*100));
xlim([10, 15]);
subplot(3,1,3);
plot(t, y_rs_ol*100, 'b--','LineWidth',0.8); hold on;
plot(t, y_rs_lqr*100, 'r', 'LineWidth',1.0);
plot(t, y_rs_hinf*100,'k', 'LineWidth',1.0);
title('Suspension Deflection');
ylabel('cm'); xlabel('Time (s)'); grid on;
legend(sprintf('OL (%.4f)', rms(y_rs_ol)*100), ...
        sprintf('LQR (%.4f)', rms(y_rs_lqr)*100), ...
        sprintf('H-inf (%.4f)',rms(y_rs_hinf)*100));
xlim([10, 15]);

% 4-8 hz analysis
% Define frequency bounds in Hz
f_lower = 4;
f_higher = 8;

% --- 1. Compute 4-8 Hz Band-Limited RMS using FFT Power Spectral Density ---
Fs = 1 / (t(2) - t(1)); % Sampling frequency
N_fft = length(t);
freq = (0:N_fft-1) * (Fs / N_fft); % Frequency vector

% Find indices corresponding to the 4-8 Hz band

```

```

band_idx = (freq >= f_lower & freq <= f_higher);

% Calculate FFTs of the body acceleration signals
fft_ol = fft(y_ab_ol) / N_fft;
fft_hinf = fft(y_ab_hinf) / N_fft;

% Extract single-sided power spectrum scaled back to RMS
% (Multiply by 2 to account for negative frequencies skipped in single-sided)
rms_4_8_ol = sqrt(sum(2 * abs(fft_ol(band_idx)).^2));
rms_4_8_hinf = sqrt(sum(2 * abs(fft_hinf(band_idx)).^2));

% Print the results nicely to the Command Window
fprintf('\n=====\\n');
fprintf('      4 - 8 Hz BANDS BODY ACCELERATION PERFORMANCE      \\n');
fprintf('=====\\n');
fprintf('  Open-Loop (Passive) RMS (4-8 Hz):  %.4f m/s^2\\n', rms_4_8_ol);
fprintf('  H-Infinity Active   RMS (4-8 Hz):  %.4f m/s^2\\n', rms_4_8_hinf);
fprintf('  Vibration Reduction Profile:        %.1f%%\\n', ...
      (rms_4_8_ol - rms_4_8_hinf) / rms_4_8_ol * 100);
fprintf('=====\\n');

% --- 2. Zoomed-In Bode Plot (4-8 Hz) for Body Acceleration ---
figure(501); clf;
opts_zoom = bodeoptions;
opts_zoom.FreqUnits = 'Hz';
opts_zoom.MagScale = 'linear'; % Linear scale is much easier to evaluate absolute di
opts_zoom.MagUnits = 'abs';

% Isolate just the 'body acceleration' channels ('ab' for H-inf, 1st output for Open-
sys_ol_acc = qcar('ab', 'r');
sys_hinf_acc = quartercar_cl_hinf('ab', 'r');

% Plot the magnitudes
bodemag(sys_ol_acc, 'b--', opts_zoom); hold on;
bodemag(sys_hinf_acc, 'k', opts_zoom);

% Set strict limits on the x-axis to zoom directly into the target zone
xlim([f_lower - 1, f_higher + 2]); % 3 Hz to 10 Hz window to see the boundaries clear
grid minor;

legend(sprintf('Open-loop (4-8Hz RMS: %.3f)', rms_4_8_ol), ...
      sprintf('H-infinity (4-8Hz RMS: %.3f)', rms_4_8_hinf), ...
      'location', 'NorthEast');
title('Zoomed Body Acceleration Frequency Response (Human Discomfort Zone)');
%% find safe speeds (no tire liftoff)
% For each ISO class, find the highest speed at which the PASSIVE car never
% lifts the tire (peak dynamic tire force < static load). Speed enters only by
% time-scaling the fixed spatial profile, so we seed the RNG to keep the road

```



```

% realisation identical while bisecting on speed.
roads_list = {'Type A','Type B','Type C','Type D','Type E','Type F','Type G','Type H'};
seed_road = 1; % fixed seed -> reproducible profiles
u_cap = 60; % upper speed bound [m/s]

u_safe = zeros(1,numel(roads_list));
fprintf('\n--- Max safe speed (no open-loop tire lift-off) ---\n');
fprintf(' road | max safe speed [m/s] | note\n');
fprintf(' -----+-----+-----\n');
for i = 1:numel(roads_list)
    [u_safe(i), lifted] = max_safe_speed_for_road(roads_list{i}, 'nom', seed_road, u_cap);
    if lifted, note = ''; else, note = sprintf('(no lift-off up to %g; capped)', u_cap);
    fprintf(' %-7s | %18.2f | %s\n', roads_list{i}, u_safe(i), note);
end

% ===== PER-ROAD AUTO-GAMMA H-INFINITY SYNTHESIS =====
% Uses synth_hinf_auto (the knob bisection, now a function) so you never set
% knob by hand: it auto-tunes knob until gamma lands near target_gamma.
beta_syn = 0.5; % comfort/handling weight for these controllers
target_gamma = 0.95;

fprintf('\n--- Per-road auto-gamma synthesis (beta = %.2f) ---\n', beta_syn);
fprintf(' road | u_safe[m/s] | knob | gamma\n');
fprintf(' -----+-----+-----+-----\n');
K_road = cell(1,numel(roads_list));
g_road = zeros(1,numel(roads_list));
knob_road = zeros(1,numel(roads_list));
for i = 1:numel(roads_list)
    [K_road{i}, g_road(i), knob_road(i)] = hinf_bisection(roads_list{i}, beta_syn, target_gamma);
    fprintf(' %-7s | %9.2f | %7.3f | %6.4f\n', ...
        roads_list{i}, u_safe(i), knob_road(i), g_road(i));
end

%% ===== PER-ROAD BETA SWEEP: 5 controllers, responses + RMS =====
% For each road class A..H: synthesise 5 H-infinity controllers with beta
% from 0.05 to 0.95 (knob auto-tuned by synth_hinf_auto), simulate each on
% that road, plot body acceleration / tire deflection / suspension deflection
% with RMS in the legend, and print an RMS table to the command window.
% NOTE: uses get_speed_for_road (should hold your 3-sigma speeds) so synthesis
% and simulation happen at the same per-road speed. ~5 controllers x 8 roads
% of auto-gamma synthesis -> this section can take a few minutes.
set(groot,'DefaultFigureWindowStyle','normal'); % undocked for this section
roads_sweep = {'Type A','Type B','Type C','Type D','Type E','Type F','Type G','Type H'};
beta_vec = linspace(0.05, 0.95, 5);
target_gamma = 0.95;
seed_sim = 1;

for ir = 1:numel(roads_sweep)
    rt = roads_sweep{ir};

```

```

u0r = get_speed_for_road(rt);

% --- road profile for this class -----
rng(seed_sim); % reproducible realisation
[~, tr, hr] = roadprofile_fun(rt, u0r);
r_iso = hr(1:numel(tr)); % road displacement
dtr = tr(2)-tr(1);
rdot = [diff(hr)./dtr, 0]; % road velocity

% --- open-loop (passive) reference -----
ab_ol = lsim(quartercar('body acceleration','r_dot'), rdot, tr);
td_ol = lsim(quartercar('tire deflection','r_dot'), rdot, tr);
sd_ol = lsim(quartercar('rattle space','r_dot'), rdot, tr);

% --- synthesise the 5 controllers and simulate -----
nb = numel(beta_vec);
AB = zeros(numel(tr),nb); TD = AB; SD = AB;
rms_ab=zeros(1,nb); rms_td=rms_ab; rms_sd=rms_ab; g_b=rms_ab;
for ib = 1:nb
    fprintf(' %s : synthesising beta=%.2f (%d/%d)...\\n', rt, beta_vec(ib), ib, nb);
    [~, g_b(ib), ~, cl] = hinf_bisection(rt, beta_vec(ib), target_gamma);
    AB(:,ib) = lsim(cl('ab','r'), r_iso, tr);
    TD(:,ib) = lsim(cl('td','r'), r_iso, tr);
    SD(:,ib) = lsim(cl('sd','r'), r_iso, tr);
    rms_ab(ib)=rms(AB(:,ib)); rms_td(ib)=rms(TD(:,ib)); rms_sd(ib)=rms(SD(:,ib));
end

% --- figure: 3 subplots, RMS in legend -----
tw0 = 0.4*tr(end); tw1 = min(tw0+5, tr(end)); % display window (edit if you like)
cols = lines(nb);
figure(100+ir); clf;
sgtitle(sprintf('Road %s | H_\\infty \\beta-sweep (u_0 = %.2f m/s): comfort \\rightarrow', rt, u0r));
subplot(3,1,1); hold on; grid on;
plot(tr, ab_ol, '--','Color',[.6 .6 .6],'DisplayName',sprintf('OL (%.3f)',rms_ab));
for ib=1:nb
    plot(tr, AB(:,ib),'Color',cols(ib,:), 'DisplayName',sprintf('\\beta=%.2f (%.3f)',beta_vec(ib),rms_ab(ib)));
end
ylabel('a_b [m/s^2]'); legend('Location','eastoutside'); xlim([tw0 tw1]);
title(sprintf('%s (u_0 = %.2f m/s) | Body Acceleration', rt, u0r));

subplot(3,1,2); hold on; grid on;
plot(tr, td_ol*100,'--','Color',[.6 .6 .6],'DisplayName',sprintf('OL (%.3f)',rms_td));
for ib=1:nb
    plot(tr, TD(:,ib)*100,'Color',cols(ib,:), 'DisplayName',sprintf('\\beta=%.2f (%.3f)',beta_vec(ib),rms_td(ib)));
end
ylabel('td [cm]'); legend('Location','eastoutside'); xlim([tw0 tw1]);
title('Tire Deflection (Road Holding)');

```



```

    rj = hj(1:numel(tj));
    rdj = [diff(hj)./(tj(2)-tj(1)), 0];
    TR(j).t=tj; TR(j).r=rj; TR(j).rdot=rdj; TR(j).u0=u0j;
    TR(j).ab_ol = lsim(quartercar('body acceleration','r_dot'), rdj, tj);
    TR(j).td_ol = lsim(quartercar('tire deflection','r_dot'), rdj, tj);
    TR(j).sd_ol = lsim(quartercar('rattle space','r_dot'), rdj, tj);
end

% --- output folder + undock figures so screenshots size correctly -----
if ~exist('figs_cross','dir'), mkdir('figs_cross'); end
prevStyle = get(groot,'DefaultFigureWindowStyle');
set(groot,'DefaultFigureWindowStyle','normal');

cmap = turbo(nR); % 8 distinct controller colours

% storage: RMS(metric).(beta){design,test}
RMSab = zeros(nR,nR,numel(betas));
RMStd = zeros(nR,nR,numel(betas));
RMSsd = zeros(nR,nR,numel(betas));

for ib = 1:numel(betas)
    beta = betas(ib);

    % ---- synthesise the 8 controllers for this beta -----
    CL = cell(1,nR);
    for i = 1:nR
        fprintf('beta=%.2f : synthesising K designed on %s (%d/%d)...\n', beta, roads{1}, nR, ib);
        [~,~,cl] = hinf_bisection(roads{i}, beta, 0.95);
        CL{i} = cl;
    end

    % ---- for each TEST road, overlay all 8 controllers -----
    for j = 1:nR
        t = TR(j).t; r = TR(j).r;
        tw0 = 0.4*t(end); tw1 = min(tw0+5, t(end));

        fig = figure(300 + (ib-1)*nR + j); clf;

        % subplot handles
        s1=subplot(3,1,1); hold on; grid on;
        s2=subplot(3,1,2); hold on; grid on;
        s3=subplot(3,1,3); hold on; grid on;

        % open-loop reference (grey dashed)
        plot(s1, t, TR(j).ab_ol, '--','Color',[.6 .6 .6],'DisplayName',sprintf('ab_ol %d',ib));
        plot(s2, t, TR(j).td_ol*100, '--','Color',[.6 .6 .6],'DisplayName',sprintf('td_ol %d',ib));
        plot(s3, t, TR(j).sd_ol*100, '--','Color',[.6 .6 .6],'DisplayName',sprintf('sd_ol %d',ib));
    end
end

```

```

for i = 1:nR
    ab = lsim(CL{i}('ab','r'), r, t);
    td = lsim(CL{i}('td','r'), r, t);
    sd = lsim(CL{i}('sd','r'), r, t);
    RMSab(i,j,ib)=rms(ab); RMStd(i,j,ib)=rms(td); RMSsd(i,j,ib)=rms(sd);

    matched = (i==j);
    lw = 2.0*matched + 1.0*~matched; % thick if design==test
    tagm = ''; if matched, tagm='*'; end
    nm_ab = sprintf('K_{%s}%s (%.3f)', letters(i), tagm, RMSab(i,j,ib));
    nm_td = sprintf('K_{%s}%s (%.3f)', letters(i), tagm, RMStd(i,j,ib)*100);
    nm_sd = sprintf('K_{%s}%s (%.3f)', letters(i), tagm, RMSsd(i,j,ib)*100);
    plot(s1, t, ab, 'Color',cmap(i,:), 'LineWidth',lw, 'DisplayName',nm_ab)
    plot(s2, t, td*100, 'Color',cmap(i,:), 'LineWidth',lw, 'DisplayName',nm_td)
    plot(s3, t, sd*100, 'Color',cmap(i,:), 'LineWidth',lw, 'DisplayName',nm_sd)
end

ylabel(s1,'a_b [m/s^2]'); ylabel(s2,'td [cm]'); ylabel(s3,'sd [cm]'); xlabel(
title(s1,'Body Acceleration'); title(s2,'Tire Deflection'); title(s3,'Suspens
xlim(s1,[tw0 tw1]); xlim(s2,[tw0 tw1]); xlim(s3,[tw0 tw1]);
legend(s1,'Location','eastoutside'); legend(s2,'Location','eastoutside'); leg
sgtitle(sprintf('Cross-test (\\beta=%.2f): controllers K_{A..H} evaluated on
        beta, letters(j), TR(j).u0));

% ---- save screenshot -----
set(fig,'Units','pixels','Position',[100 100 1300 950]); drawnow;
fname = sprintf('crosstest_%s_Road%s', btag{ib}, letters(j));
exportgraphics(fig, fullfile('figs_cross',[fname '.png']), 'Resolution', 200)
fprintf(' saved: %s.png\\n', fname);
end

% ---- command-window cross matrices (rows=design, cols=test) -----
print_cross(sprintf('beta=%.2f BODY-ACC RMS [m/s^2]',beta), RMSab(:,:,ib), let
print_cross(sprintf('beta=%.2f TIRE-DEFL RMS [cm]', beta), RMStd(:,:,ib)*100,le
print_cross(sprintf('beta=%.2f SUSP-DEFL RMS [cm]', beta), RMSsd(:,:,ib)*100, le
end

set(groot,'DefaultFigureWindowStyle',prevStyle); % restore docking pref
% ===== end of cross-test section =====

%% helpers
function print_cross(ttl, M, letters, ~)
    nR = numel(letters);
    fprintf('\\n=== %s (rows = design road, cols = test road) ===\\n', ttl);
    fprintf(' design\\test |');
    for j=1:nR, fprintf('%8s', letters(j)); end
    fprintf('\\n -----+');
    for j=1:nR, fprintf('-----'); end

```

```

fprintf('\n');
for i=1:nR
    fprintf('      K_%s      |', letters(i));
    for j=1:nR
        if i==j, fprintf('%7.3f*', M(i,j)); else, fprintf('%8.3f', M(i,j)); end
    end
    fprintf('\n');
end
fprintf(' (* = matched: controller tested on its own design road)\n');
end

function [u_safe, lifted] = max_safe_speed_for_road(road_type, params, seed, u_cap)
% Highest speed at which the passive car does NOT lift the tire.
    if nargin<2||isempty(params), params='nom'; end
    if nargin<3||isempty(seed), seed=1; end
    if nargin<4||isempty(u_cap), u_cap=60; end

    [ms,mu,bs,ks,kt,bt] = get_vehicle_params(params);
    g = 9.81; F_static = (ms+mu)*g;
    Acar = [0 1 0 -1; -ks/ms -bs/ms 0 bs/ms; 0 0 0 1; ks/mu bs/mu -kt/mu -(bs+bt)/mu]
    L = [0;0;-1;0]; Ctd = [0 0 1 0];
    Gtd = ss(Acar, L, Ctd, 0); % road velocity -> tire deflection (passive)

    % no lift-off even at the cap -> return cap
    if ~road_lifts_off(u_cap, road_type, seed, Gtd, kt, F_static)
        u_safe = u_cap; lifted = false; return;
    end
    % lift-off already at the lowest speed -> return that
    u_lo = .1; u_hi = u_cap;
    if road_lifts_off(u_lo, road_type, seed, Gtd, kt, F_static)
        u_safe = u_lo; lifted = true; return;
    end
    % bisection: u_lo is safe, u_hi lifts
    for it = 1:22
        u_mid = (u_lo+u_hi)/2;
        if road_lifts_off(u_mid, road_type, seed, Gtd, kt, F_static)
            u_hi = u_mid;
        else
            u_lo = u_mid;
        end
    end
    u_safe = u_lo; lifted = true;
end

function tf = road_lifts_off(u0, road_type, seed, Gtd, kt, F_static)
    rng(seed); % same spatial profile every call
    [~, t, hsum] = roadprofile_fun(road_type, u0);
    dt = t(2)-t(1);

```

```

    rdot = [diff(hsum)./dt, 0]; % road velocity (scales with u0)
    td = lsim(Gtd, rdot, t);
    % tf = max(abs(kt*td)) >= F_static;
    sigma_F = std(kt*td); % std of dynamic tire force
    tf = 3*sigma_F >= F_static; % 3-sigma road-holding limit
end

function [K, gamma_out, knob, cl_hinf, qcar_open] = hinf_bisection(road_type, beta, t
% Knob bisection (your old inline loop, now reusable): auto-tunes knob until
% gamma is near target_gamma, returning the controller at that knob.
    if nargin<3||isempty(target_gamma), target_gamma = 0.95; end
    if nargin<4||isempty(params), params = 'nom'; end
    gamma_tol = 0.05; max_iters = 50;
    knob_low = 0.01; knob_high = 50.0;
    K = []; gamma_out = Inf; knob = 1; cl_hinf = []; qcar_open = [];

    for it = 1:max_iters
        knob_test = (knob_low + knob_high)/2;
        [cl, Kt, gt, qc] = design_hinf_for_road(road_type, 1, 1, knob_test, beta, par

        if isempty(gt) || isnan(gt) || isinf(gt)
            knob_high = knob_test; continue; % infeasible -> soften weights
        end
        K = Kt; gamma_out = gt; knob = knob_test; cl_hinf = cl; qcar_open = qc;

        if abs(gt - target_gamma) <= gamma_tol, break; end
        if gt > target_gamma, knob_high = knob_test; else, knob_low = knob_test; end
    end
end

function [cl_hinf, K, gamma_out, qcar_open, info, rdot_iso, r_iso] = design_hinf_for_

    if nargin < 6 || isempty(params)
        params = 'nom';
    end

    u0 = get_speed_for_road(road_type); V = u0;
    [~, t, hsum, Gh0] = roadprofile_fun(road_type, u0);
    rms_road = rms(hsum);
    r_iso = hsum(1:length(t));
    dt = t(2) - t(1);
    rdot_iso = [diff(hsum)./dt, 0];

    [ms, mu, bs, ks, kt, bt] = get_vehicle_params(params);

    % --- Displacement-input plant for H-infinity synthesis ---
    A = [ 0 1 0 0; ...
        [-ks -bs ks bs]/ms; ...

```

```

0 0 0 1; ...
[ks bs -ks-kt -bs]/mu];
B = [ 0 0; 0 1e3/ms; 0 0; [kt -1e3]/mu];

% addition (tire deflection): td = x3 - r
C = [1 0 0 0; 1 0 -1 0; A(2,:); 0 0 1 0];
D = [0 0; 0 0; B(2,:); -1 0];

qcar_open = ss(A, B, C, D);
qcar_open.StateName = ["body travel (m)"; "body vel (m/s)"; ...
    "wheel travel (m)"; "wheel vel (m/s)"];
qcar_open.InputName = ["r"; "fs"];
qcar_open.OutputName = ["xb"; "sd"; "ab"; "td"];

w_bounce = sqrt(ks/ms); % body bounce freq
w_wheel_hop = sqrt(kt/mu); % wheel-hop freq

Act = tf(1, [1/60 1]); % actuator with 60 rad/s cutoff, light but not unity
Act.InputName = "u"; Act.OutputName = "fs";

w_road_cut = 2 * pi * 2; % cut-off frequency = 2 Hz
Wroad = tf(sqrt(Gh0 * V) * w_road_cut, [1, w_road_cut]);
Wroad.u = "d1"; Wroad.y = "r";

Wact = 0.1 * tf([1, w_bounce], [1, 100]);
Wact.u = "u"; Wact.y = "e1";

Wd2 = ss(0.01); Wd2.u = "d2"; Wd2.y = "Wd2"; % measurement noise, small
Wd3 = ss(0.01); Wd3.u = "d3"; Wd3.y = "Wd3"; % measurement noise, small

max_travel = 1.0; % for know it's 1 meters, will change
scale_factor = 1/max_travel;
Wsd = scale_factor * tf(1, [1/w_bounce, 1]);
Wsd.u = "sd"; Wsd.y = "e3";

% human are sensitive to 4-8 Hz (25.1-50.3 rad)
w_lower = 25.1; w_higher = 50.3;
w_center = (w_lower + w_higher)/2;
wab_bandwidth = w_higher-w_lower;
Wab = knob * (1-beta) * tf([wab_bandwidth, 0], [1, wab_bandwidth, w_center^2]);
Wab.u = "ab"; Wab.y = "e2";

% Road handling is evaluated at the wheel-hop frequency
% Wtd = knob * beta * tf(1, [1/w_wheel_hop, 1]);
Wtd = knob * beta * tf([1, 1], [1, w_wheel_hop]);
Wtd.u = "td"; Wtd.y = "e4";

% make necessary connections for plant

```



```

sdmeas = sumblk("y1 = sd+Wd2");
abmeas = sumblk("y2 = ab+Wd3");
ICinputs = ["d1";"d2";"d3";"u"];
ICoutputs = ["e1";"e2";"e3";"e4";"y1";"y2"];

% create plant
qcaric = connect(qcar_open(["sd","ab","td"],:), Act, Wroad, Wact, Wab, Wsd, Wtd,
                Wd2, Wd3, sdmeas, abmeas, ICinputs, ICoutputs);

[K, ~, gamma_out] = hinfsyn(qcaric, 2, 1); % synthesize hinf controller
K.u = ["sd","ab"]; K.y = "u";

% Reconnect closed loop with ALL outputs including Tire Deflection
cl_hinf = connect(qcar_open, Act, K, "r", ["xb";"sd";"ab";"td"]);
cl_hinf = cl_hinf([3,2,1,4], :); % Orders outputs as: ab, sd, xb, td

info.road_type = road_type;
info.u0 = u0;
info.t = t;
info.r_iso = r_iso;
info.hsum = hsum;
info.rms_road = rms_road;
info.rms_a_ref = rms_a_ref;
info.rms_rs_ref = rms_rs_ref;
info.ms = ms; info.mu = mu; info.ks = ks; info.kt = kt;
info.bs = bs; info.bt = bt;
end

function u0 = get_speed_for_road(road_type)
    switch road_type
        case 'Type A', u0 = 60.00; % 3-sigma no-lift-off (capped)
        case 'Type B', u0 = 60.00; % 3-sigma no-lift-off (capped)
        case 'Type C', u0 = 33.36;
        case 'Type D', u0 = 9.00;
        case 'Type E', u0 = 4.10;
        case 'Type F', u0 = 2.97;
        case 'Type G', u0 = 1.46;
        case 'Type H', u0 = 0.57;
        otherwise
            error('Unknown road type: "%s"', road_type);
        end
    end
end

function [ms, mu, bs, ks, kt, bt] = get_vehicle_params(params)
    switch params
        case 'bd'
            ms = 300; mu = 60; bs = 1000; ks = 16000; kt = 190000; bt = 0;
        case 'nom'

```

```

        ms = 240; mu = 36;  bs = 1000; ks = 16000;  kt = 160000; bt = 0;
case 'suv'
    ms = 600; mu = 60;  bs = 3000; ks = 20000;  kt = 180000; bt = 0;
case 'truck'
    ms = 4000; mu = 200; bs = 8000; ks = 150000; kt = 800000; bt = 0;
case 'sunlusoy'
    ms = 250; mu = 35;  bs = 800;  ks = 15000;  kt = 120000; bt = 0;
otherwise
    error('Unknown params: "%s"', params);
end
end

```